Foundations of Modern Networking

SDN, NFV, QoE, IoT, and Cloud

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Chapter 1

Elements of Modern Networking

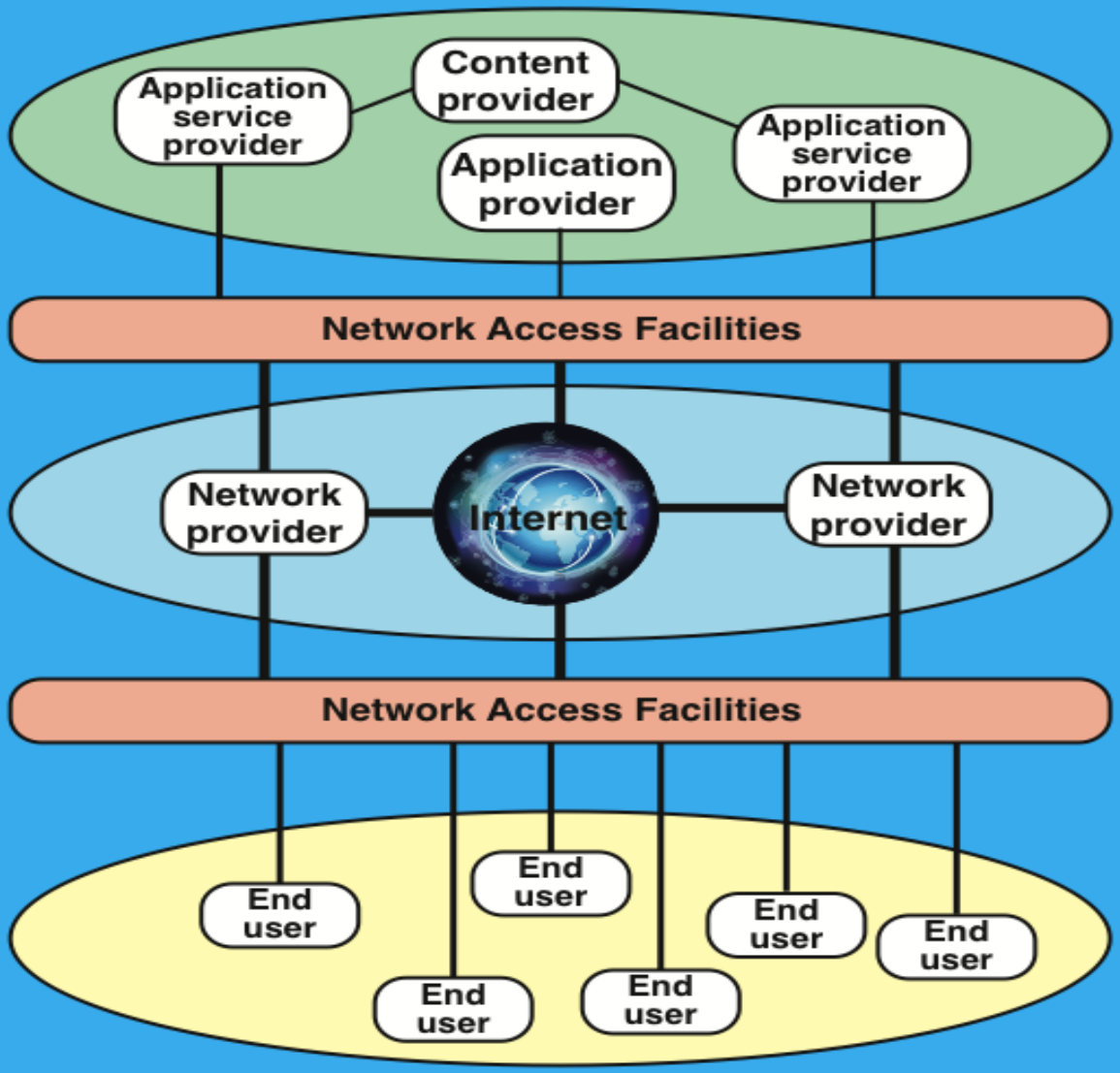


Figure 1.1 The Modern Networking Ecosystem

digital subscriber line (DSL) and cable modems, Wi-Fi and Worldwide Interoperability for Microwave Access (WiMAX) wireless modems, and cellular modems

Here two major elements of modern networking not explicitly depicted in this previous figure:

- Data center networking: Both large enterprise data centers and cloud provider data centers consist of very large numbers of interconnected servers. Typically, as much as 80 percent of the data traffic is within the data center network, and only 20 percent relies on external networks to reach users.

- IoT or fog networking: An Internet of Things deployed by an enterprise may consist of hundreds, thousands, even millions of devices. The vast bulk of the data traffic to and from these devices is machine to machine, rather than user to machine.

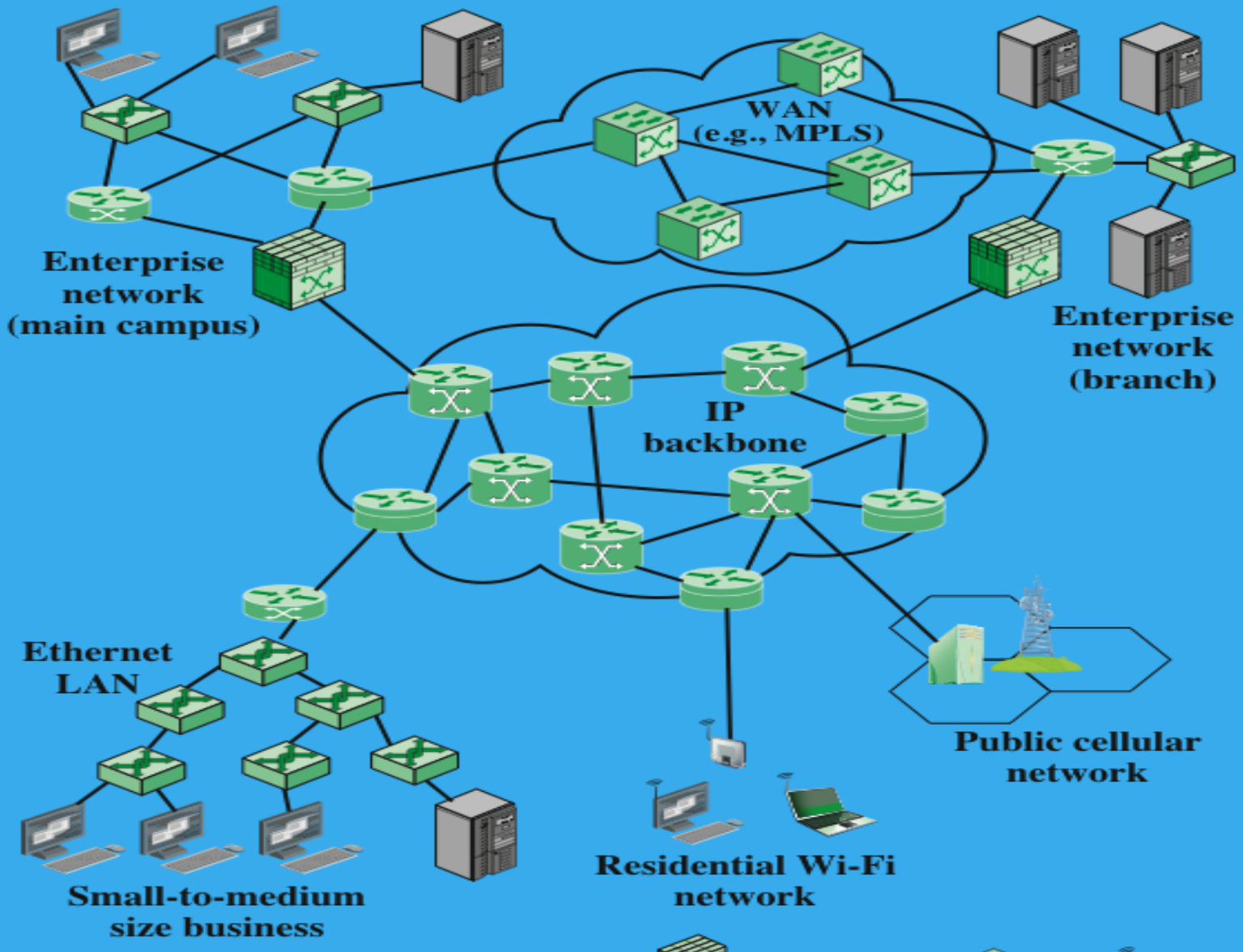


Figure 1.2 A Global Network Architecture

- Typically, the IP backbone consists of high-performance routers, called core routers, interconnected with high-volume optical links.
  - The optical links often make use of wavelength division multiplexing (WDM), such that each link has multiple logical channels.
- Edge routers or aggregation routers that provide connectivity to external networks and users.
  - Aggregation routers are also used within an enterprise network to connect a number of routers and switches, to external resources, such as an IP backbone or a high-speed WAN.
- The analysis concludes that aggregation router requirements will be in the range of 200 Gbps to 400 Gbps per optical link by 2020, and 400 Gbps to 1 Tbps per optical link for core routers by 2020.

- Two sections of a large enterprise network connected via a private highspeed WAN, with switches interconnected with optical links.
  - MPLS using IP is a common switching protocol used for such WANs; wide-area Ethernet is another option.
  - Enterprise assets are connected to, and protected from, an IP backbone or the Internet via routers with firewall capability

- A small- or medium-size business, which relies on an Ethernet LAN.
  - Connection to the Internet through a router could be through a cable or DSL connection or a dedicated high-speed link.

- An individual residential user connected to an Internet service provider (ISP) through some sort of subscriber connection.
  - Common examples:
    - a DSL, which provides a high-speed link over telephone lines and requires a special DSL modem
    - a cable TV facility, which requires a cable modem, or some type of wireless connection.

- Finally, mobile devices, such as smartphones and tablets, can connect to the Internet through the public cellular network, which has a high-speed connection, typically optical, to the Internet.

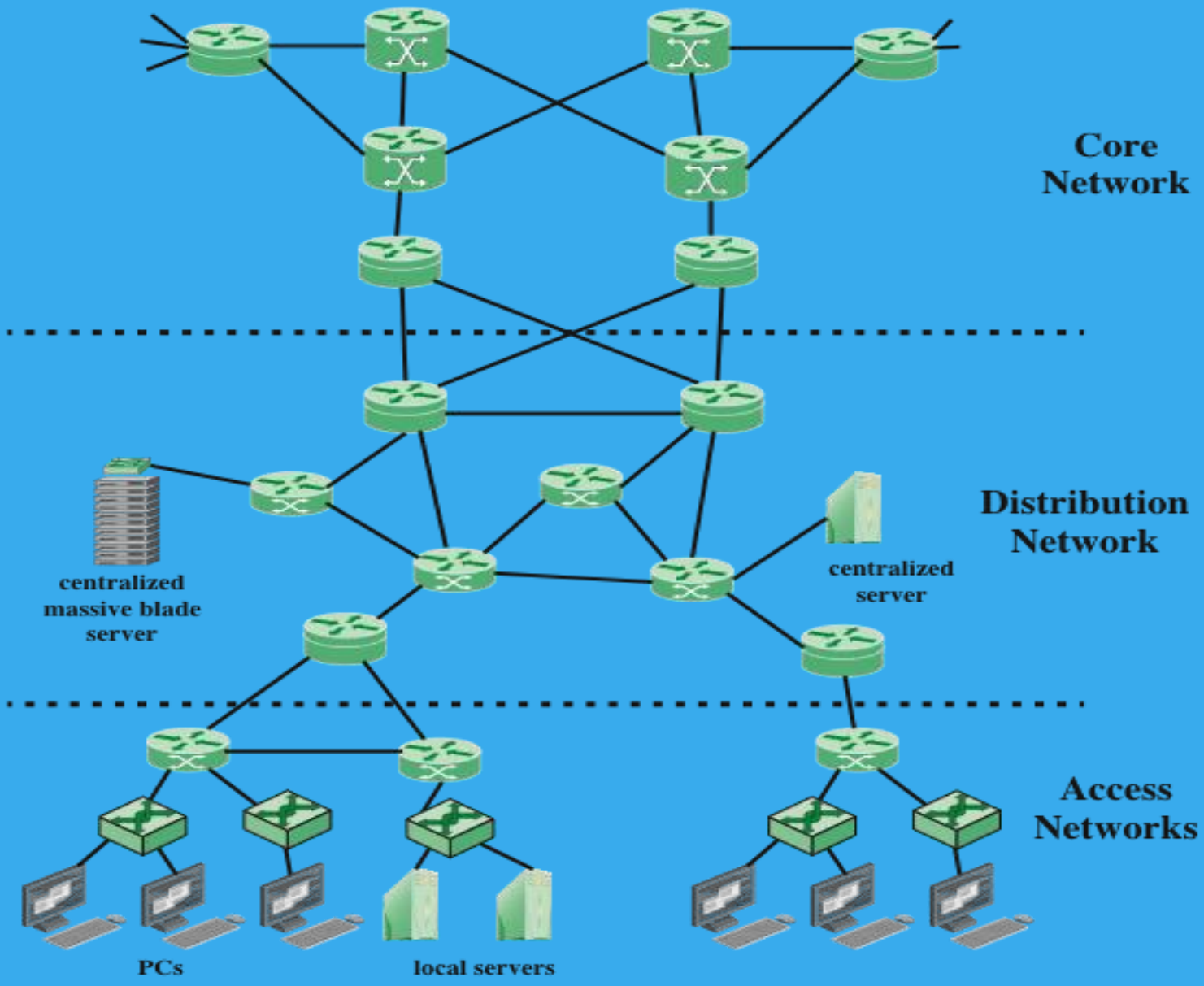


Figure 1.3 A Typical Network Hierarchy

- Closest to the end user is the access network.

  - Typically, an access network is a local-area network (LAN) or campus-wide network that consisting of LAN switches(typically Ethernet switches) and, in larger LANs, IP routers that provide connectivity among the switches.

  - The access network supports end user equipment, such as desktop and laptop computers and mobile devices.

  - The access network also supports local servers that primarily or exclusively serve the users on the local access network.

- One or more access routers act as edge routers connect the distribution network via the Internet or some other public or private communications facility.

  - these access routers that forward traffic into and out of the access network.

  - For a large local facility, there might be additional access routers that provide internal routing but do not function as edge routers.

- An edge router in the distribution network connects to an edge router in an access network to provide connectivity.
  - The two routers are configured to recognize each other and will generally exchange routing and connectivity information and, typically, some traffic-related information. This cooperation between routers is referred to as peering.

  - The distribution network also serves to aggregate traffic destined for the core router, which protects the core from high-density peering.
  - A distribution network may also directly connect servers that are of use to multiple access networks, such as database servers and network management servers.

- Again, as with access networks, some of the distribution routers may be purely internal and do not provide an edge router function.

- The core network, also referred to as a backbone network , connects geographically dispersed distribution networks as well as providing access to other networks that are not part of the enterprise network.
  - Typically, the core network will use very high performance routers, high-capacity transmission lines, and multiple interconnected routers for increased redundancy and capacity.
  - The core network may also connect to high-performance, high-capacity servers, such as large database servers and private cloud facilities.
  - Some of the core routers may be purely internal, providing redundancy and additional capacity without serving as edge routers.

- A hierarchical network architecture is an example of a good modular design.
  - With this design, the capacity, features, and functionality of network equipment (routers, switches, network management servers) can be optimized for their position in the hierarchy and the requirements at a given hierarchical level.

Ethernet

- Predominant wired networking technology
- Has evolved to support data rates up to 100Gbps
- Supports distances from a few meters to tens of kilometers
- Has become essential for supporting personal computers, workstations, servers, and massive data storage devices in organizations large and small

Ethernet in the Home

- Has been used to create a local network of computers with access to the Internet via a broadband modem/router
- Home reliance on Ethernet has declined with increasing availability of high-speed, low-cost Wi-Fi

Ethernet in the Home

- Two recent extensions of Ethernet technology have enhanced and broadened the use of Ethernet in the home
  - Powerline carrier (PLC)
    - existing power lines and use the power wire as a communication channel to transmit Ethernet packets on top of the power signal
  - Power over Ethernet (PoE)
    - distributing power over the Ethernet data cable
    - distribute power to devices on the network
    - simplifying the wiring for devices

Ethernet in the Office

- Has been the dominant network technology for wired local-area networks (LANs) in the office environment
- competitors, such as IBM's Token Ring LAN and the Fiber Distributed Data Interface (FDDI), but the simplicity, performance, and wide availability of Ethernet hardware eventually made Ethernet the winner

Retains its popularity because:

- It can support many devices at high speeds
- Is not subject to interference
- Provides a security advantage because it is resistant to eavesdropping
- A combination of Ethernet and Wi-Fi is the most common architecture

- The LAN connects to the Internet/WANs via a firewall.

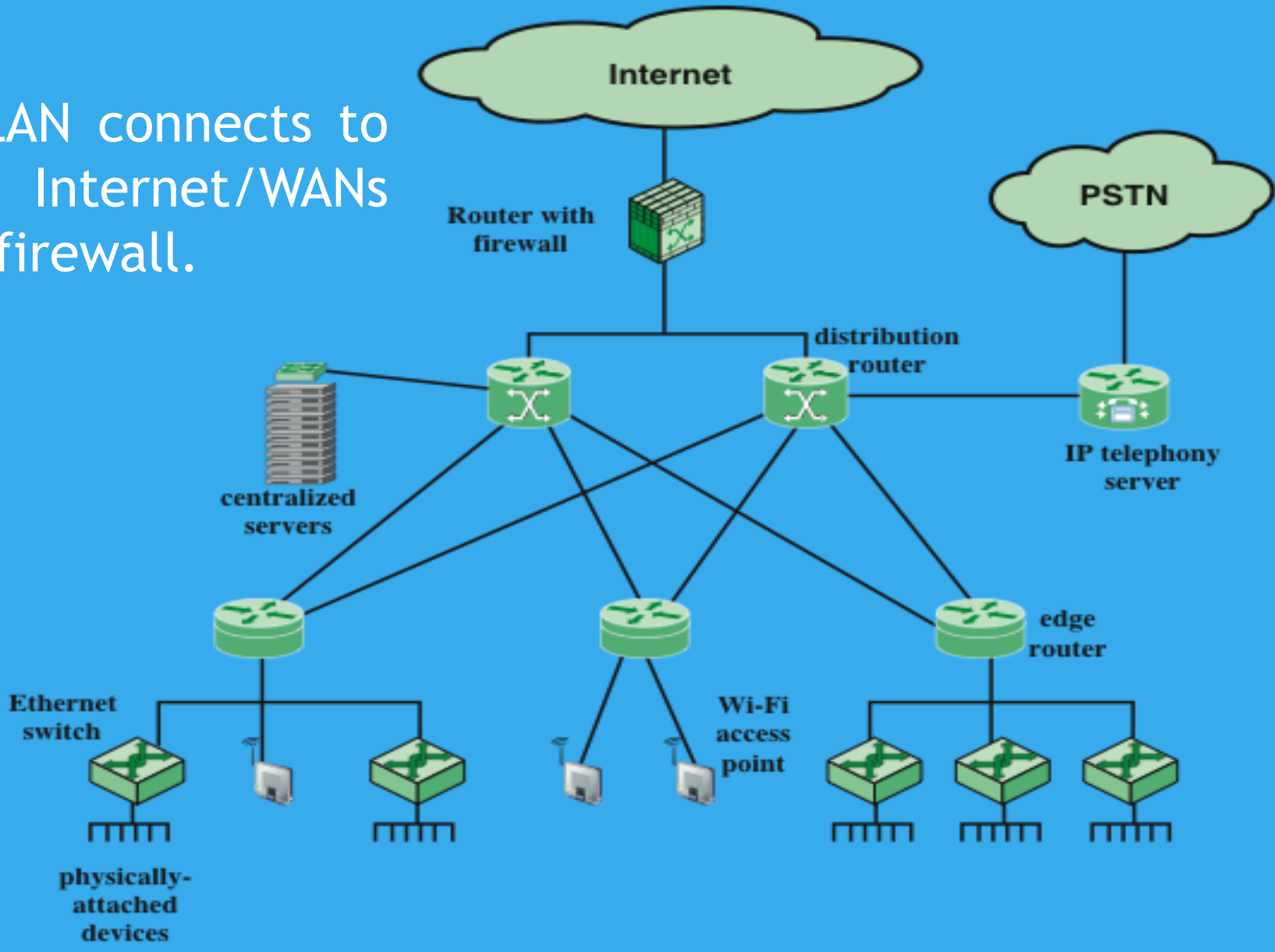
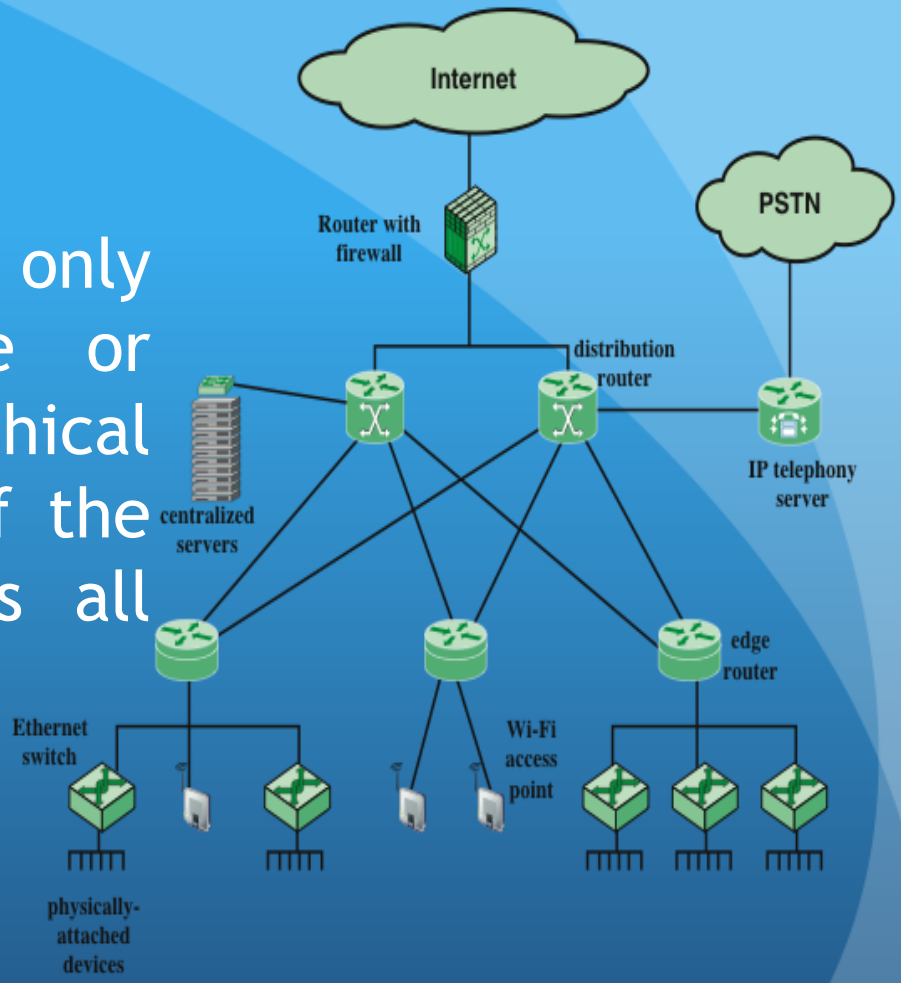
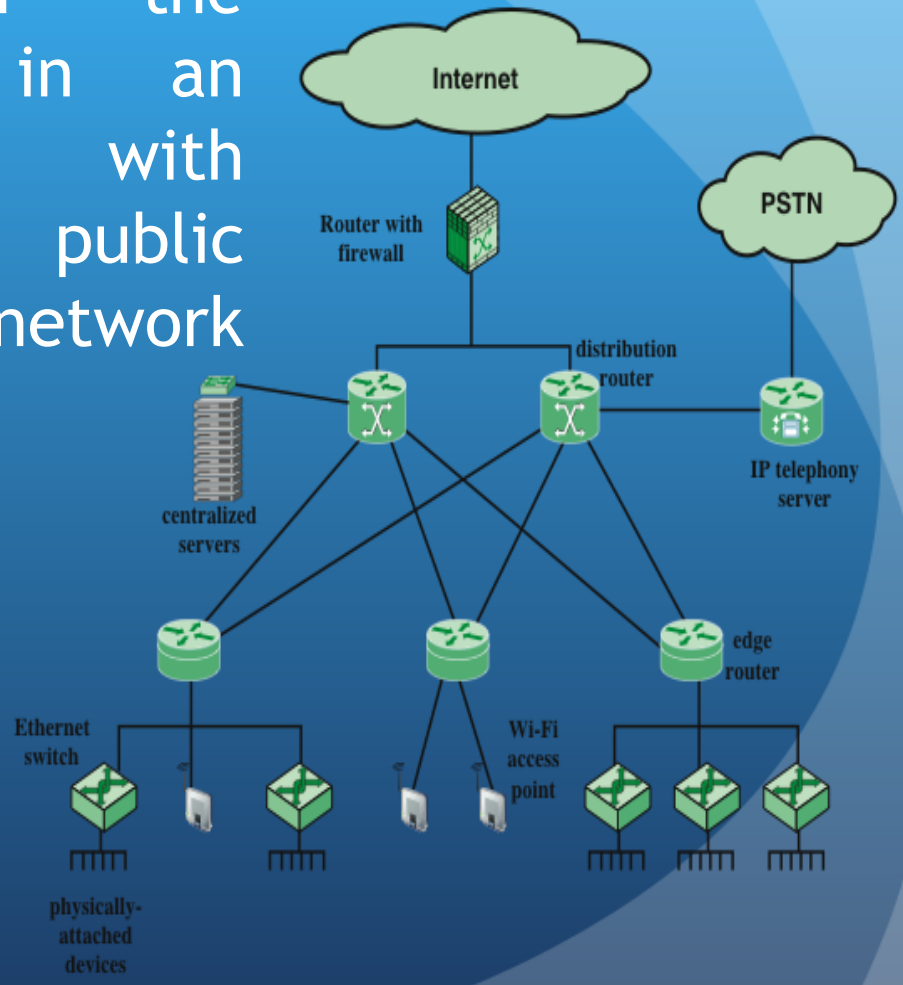


Figure 1.4 A Basic Enterprise LAN Architecture

- Wireless devices are only attached at the edge or bottom of the hierarchical architecture; the rest of the campus infrastructure is all Ethernet.



- An IP telephony server that provides call control functions (voice switching) for the telephony operations in an enterprise network, with connectivity to the public switched telephone network (PTSN).



Ethernet in the Enterprise

- Tremendous advantage of Ethernet is that it is possible to scale the network, both in terms of distance and data rate with the same Ethernet protocol and associated quality of service
- An enterprise can easily extend an Ethernet network using a mixture of cable types and Ethernet hardware
- Because all the hardware and communications software conform to the same standard it is easy to mix different speeds and different vendor equipment
  - The same protocol is used for intensive high-speed interconnections of data servers in a single room

Ethernet in the Data Center

- Has come to dominate in the data center where very high data rates are needed to handle massive volumes of data among networked servers and storage units
- For co-located servers and storage units, high-speed Ethernet fiber links and switches provided the needed networking infrastructure
- Backplane Ethernet
  - Runs over copper jumper cables that can provide up to 100 Gbps over very short distances
- Blade servers
  - Multiple server modules housed in a single chassis

Ethernet for Wide-Area Networking

- Until recently Ethernet was not a significant factor in wide-area networking, Gradually more telecommunications and network providers have switched to Ethernet from alternative schemes
- Other wide-area options
  - Dedicated T1 lines
    - Special telephone lines for voice and internet traffic
  - Synchronous digital hierarchy (SDH) lines
    - High-speed digital lines for long distance communication
  - Asynchronous Transfer Mode (ATM)
    - Complex system for carrying voice, video and data data
- Carrier Ethernet
  - One of the fastest-growing Ethernet technologies

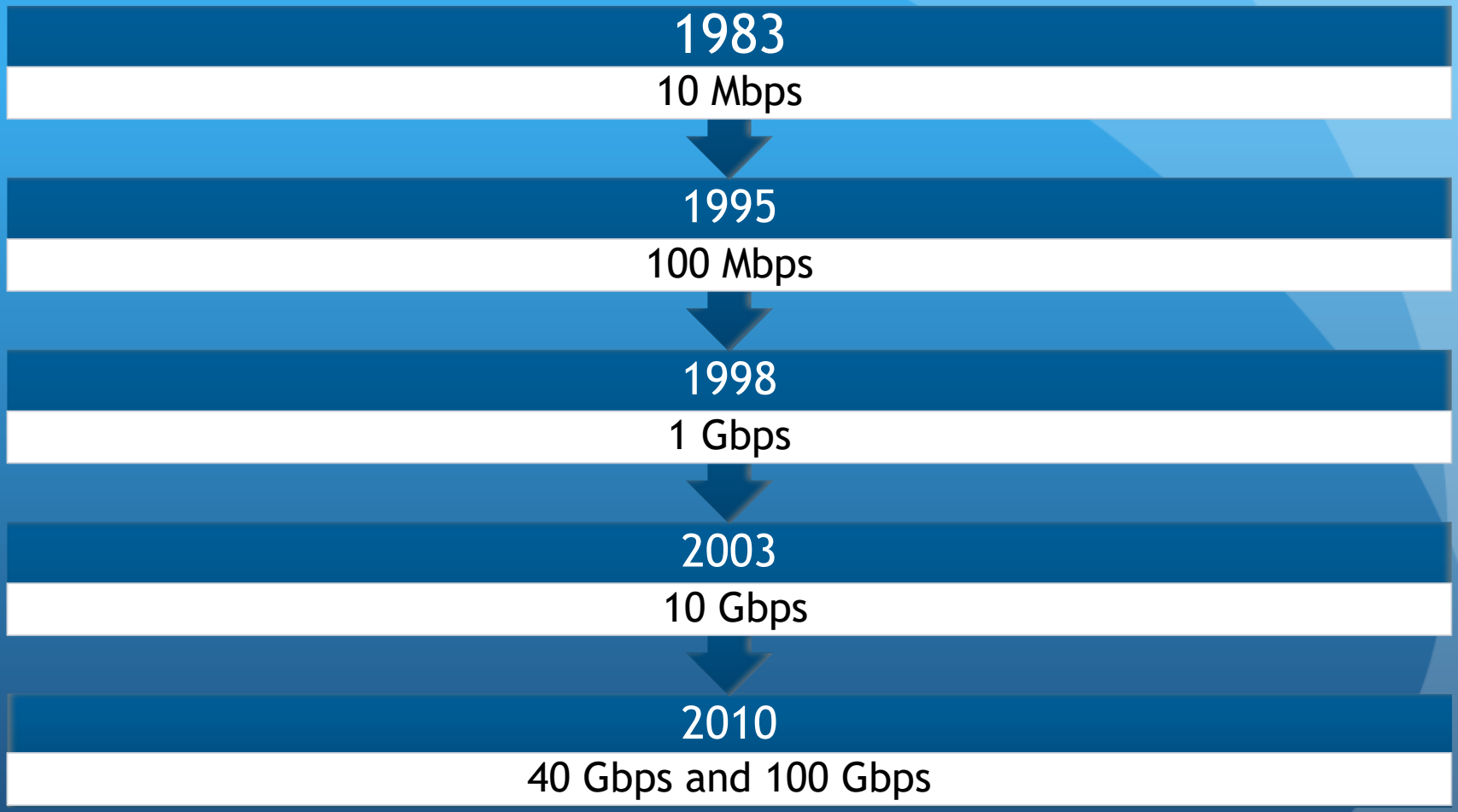
Standards

IEEE 802 LAN standards committee

- 802.3 group is responsible for issuing standards for LANs that are referred to commercially as Ethernet

The Ethernet Alliance supports and originates activities that span from incubation of new Ethernet technologies to interoperability testing to demonstrations to education

Ethernet Data Rates



- Currently Ethernet systems are available at speeds up to 100 Gbps

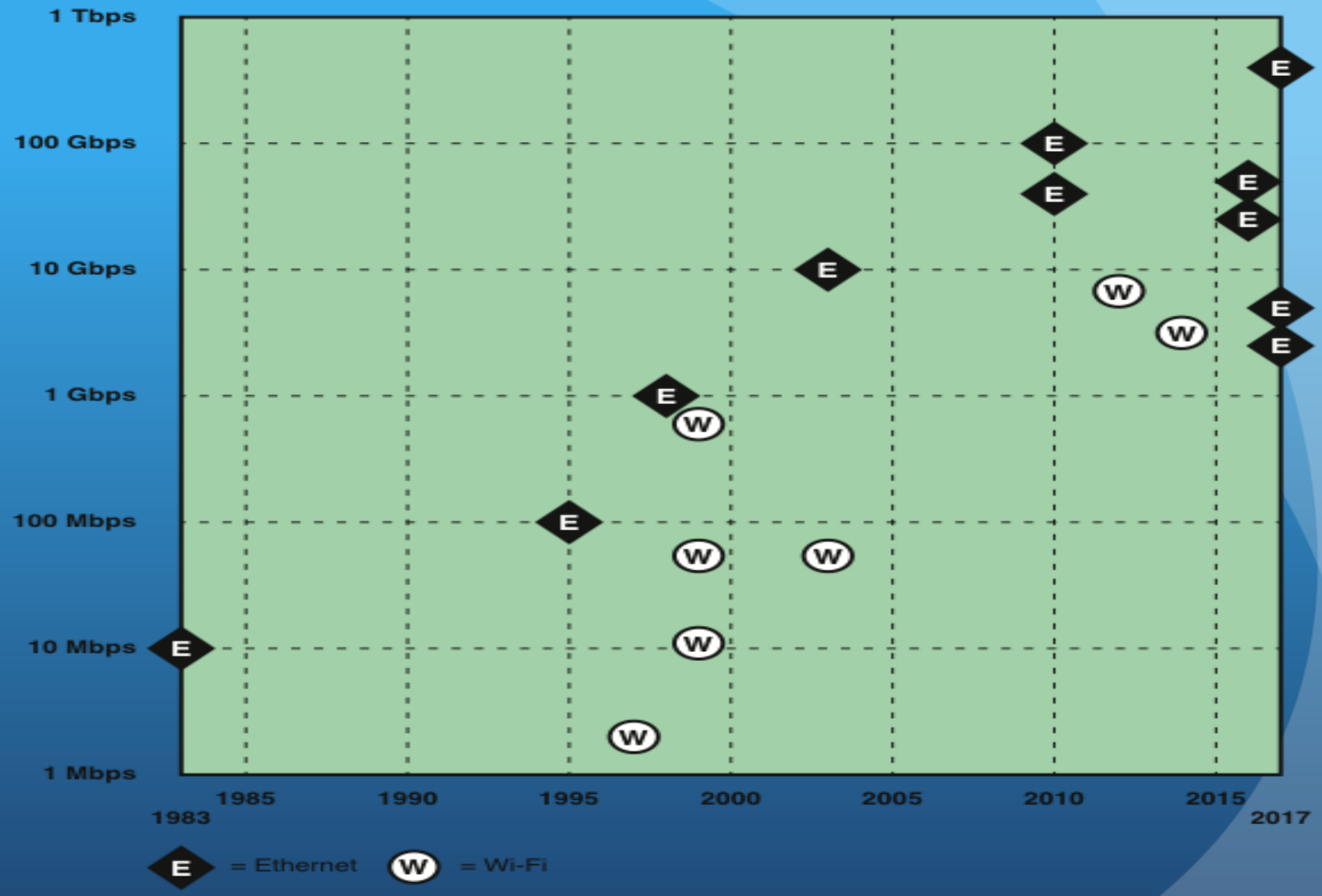


Figure 1.5 Ethernet and Wi-Fi Timelines

1-Gbps Ethernet

Centralized server farms

- In many multimedia applications there is a need for client systems to be able to draw huge amounts of data from multiple centralized servers, called server farms
- Server performance increased

High-speed local backbone

- As processing demand grows, enterprises develop an architecture of multiple LANs interconnected with a high-speed backbone network

Power workgroups

- These groups typically consist of a small number of cooperating users who need to exchange massive data files across the network
- Example applications are software development and computer-aided design

10-Gbps Ethernet

- The principle driving requirement for 10-Gbps Ethernet was the increase in intranet and Internet traffic
- Factors contributing to the explosive growth in both Internet and intranet traffic:
  - An increase in the number of network connections
  - An increase in the connection speed of each end-station
  - An increase in the deployment of bandwidth-intensive applications such as high-quality video
  - An increase in web hosting and application hosting traffic
- Initially to provide high-speed, local backbone interconnection between large-capacity switches
  - Due to demand, deployed throughout the networks, server farms, backbone, and campus-wide connectivity

100-Gbps Ethernet

- Market drivers for 100-Gbps Ethernet
  - Data center/Internet media providers
    - To support the growth of Internet multimedia content and web applications, content providers have been expanding data centers
  - Metro video/service providers
    - Video on demand has been driving a new generation of 10-Gbps Ethernet

100-Gbps Ethernet

- Market drivers for 100-Gbps Ethernet
  - Enterprise LANs
    - Continuing growth in convergence of voice, video, and data and in unified communications is driving up network switch demands
  - Internet exchanges/ISP core routing
    - With the massive amount of traffic flowing through these nodes, these installations are likely to be early adopters of 100-Gbps Ethernet

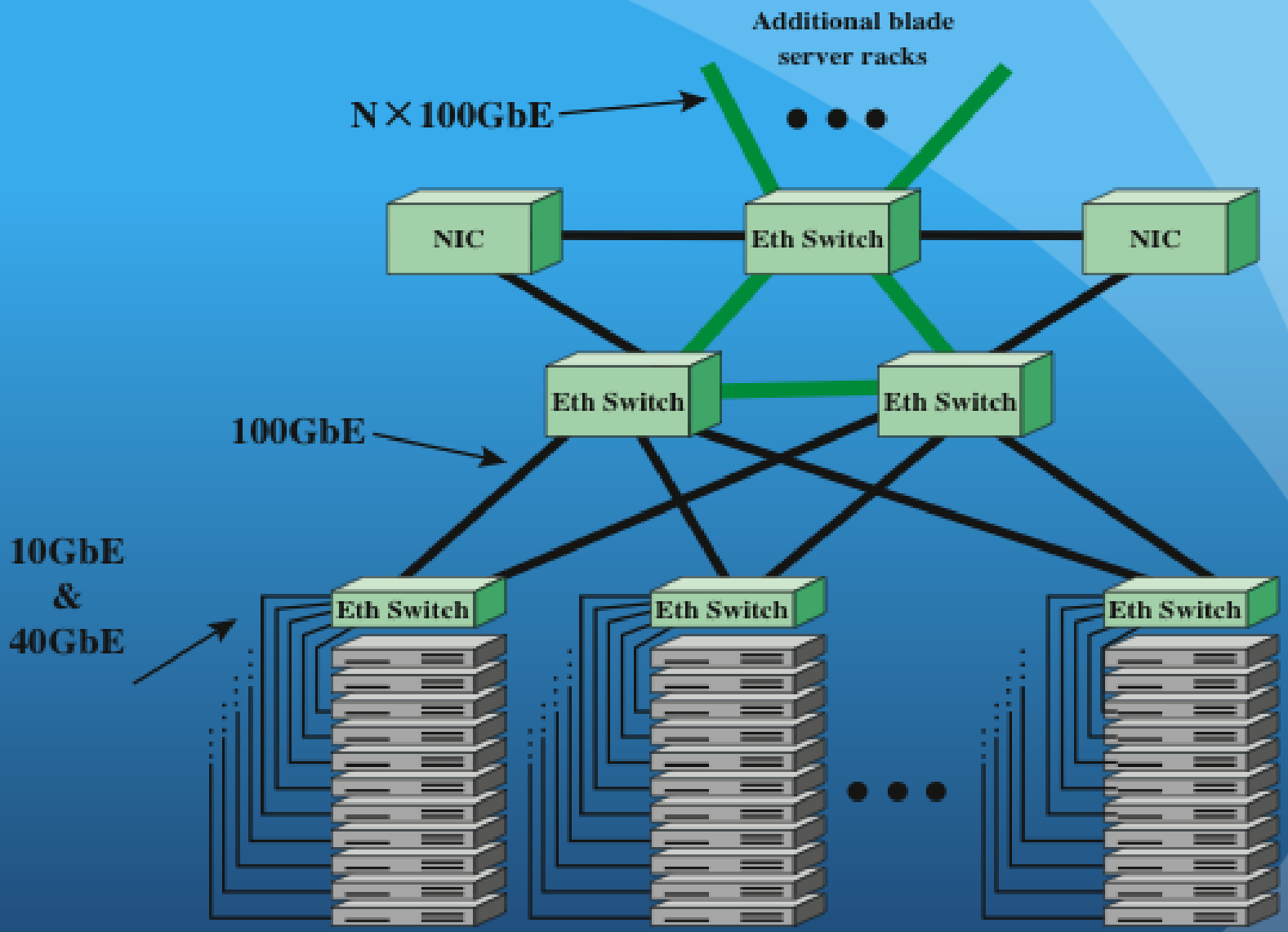


Figure 1.6 Configuration for Massive Blade Server Cloud Site

25/50 Gbps

- 25 Gigabit Ethernet Consortium
  - Includes cloud networking providers such as Google and Microsoft
  - Objective is to support an industry standard, interoperable Ethernet specification that boosts the performance and slashes the interconnect cost per Gbps between the NIC and ToR switch
  - Prescribes a single-lane 25-Gbps Ethernet and dual-lane 50 Gbps Ethernet link protocol
  - IEEE 802.3 committee is at work developing the needed standards for 25 Gbps and may include 50 Gbps

400 Gbps Ethernet

- IEEE 802.3 is currently exploring technology options for producing a 400-Gbps/800-Gbps Ethernet standard
- 1-Tbps (terabits per second, trillion per second) standard will eventually be produced
- No timetable is yet in place

2.5/5 Gbps Ethernet

- These lower speeds are known as Multirate Gigabit BASE-T (MGBASE-T)
- Currently the MGBASE-T Alliance is overseeing the development of these standards outside of IEEE
- Mainly intended to support IEEE 802.11ac wireless traffic into a wired network
- Gaining acceptance where more than 1 Gbps of throughput is needed (such as to support mobile users in the office environment)

Wi-Fi

- Standardized by IEEE 802.11
- Has become the dominant technology for wireless LANs
- First important use was in the home to replace Ethernet cabling for connecting desktop and laptop computers with each other and with the Internet
- Provides a cost effective way to the Internet
- Is essential to implementing the Internet of Things (IoT)

Public Wi-Fi

Hotspots have become readily available

Remote places can support hotspots with the development of the satellite Wi-Fi hotspot

- First company to develop such a product was Iridium, now we have starlink

The satellite modem will initially provide a relatively low-speed connection, but the data rates will inevitable increase

Enterprise Wi-Fi

- Economic benefit of Wi-Fi is most clearly seen in the enterprise
- Approximately half of all enterprise network traffic is via Wi-Fi rather then the traditional Ethernet
- Two trends have driven the transition to a Wi-Fi centered enterprise:
  - Demand has increased with more and more employees preferring to use laptops, tablets, and smartphones to connect to the enterprise network
  - The arrival of Gigabit Ethernet allows the enterprise network to support high-speed connections to mobile devices simultaneously

Standards

- Interoperability is essential to the success of Wi-Fi
- Wi-Fi enabled devices must be able to communicate with Wi-Fi access points regardless of the manufacturer of the device or access point
- Interoperability is guaranteed by:
  - IEEE 802.11 wireless LAN committee develops the protocol and signaling standards
  - The Wi-Fi Alliance creates test suites to clarify interoperability for commercial products that conform to various IEEE 802.11 standards
  - The term *Wi-Fi* (wireless fidelity) is used for products certified by the Alliance

Wi-Fi Data Rates

- 802.11 (1997): 2 Mbps (megabits per second, million bits per second), 2.4 GHz band
- 802.11a (1999): 54 Mbps, 5 GHz band
- 802.11b (1999): 11 Mbps, 2.4 GHz band
- 802.11g (2003): 54 Mbps, 2.4 GHz
- 802.11n (2009): 600 Mbps, 2.4/5 GHz, MIMO

Wi-Fi Data Rates

- 802.11ad (2012): 6.76 Gbps (billion bits per second)
  - operating in the 60-GHz frequency band
  - Few devices operate in the 60-GHz band, which means communication experiences less interference
  - Useful within a single room
  - Official and residential enviroments
  - High data rates, easily transmit uncompressed high-definition video

Wi-Fi Data Rates

- 802.11ac (2014): 3.2 Gbps
  - operates in the 5-GHz band
  - makes use of advanced technologies in antenna design and signal processing to achieve much greater data rates, at lower battery consumption
- 802.11ax (2019): 9.6 Gbps, OFDMA, multi-user MIMO in both downlink and uplink
- 802.11be (2024): 46 Gbps, supports 2.4 GHz, 5 GHz, and 6 GHz bands.

4G/5G Cellular

First Generation — 1G

- Original cellular networks
- Provided analog traffic channels and were designed to be an extension of the public switched telephone networks
- Most widely deployed system was the Advance Mobile Phone Service (AMPS) developed by AT&T
- Voice transmission was purely analog and control signals were sent over a 10-kbps analog channel
- FDMA is used

4G/5G Cellular

Second Generation 2G

- Developed to provide:
  - Higher-quality signals
  - Higher data rates for support of digital services
  - Greater capacity

4G/5G Cellular

Second Generation 2G

- Key differences:
  - Digital traffic channels
    - 2G systems are digital
    - 1G was to support voice signals, need a modem to convert to digital data into analog signal
    - Voice is encoded before transmitting
  - Encryption
    - User and control traffic is digitized, easy to encrypt the traffic to prevent eavedropping

4G/5G Cellular

Second Generation 2G

- Key differences:
  - Error detection and correction
    - Use of error detection and correction techniques, result is clear voice reception
  - Channel access
    - In 1G, each cell supports a number of channels
      - A channel is allocated to only one user
    - 2G system also provide multiple channels per cell
      - Each channel is dynamically shared by a number users
    - TDMA +FDMA

4G/5G Cellular

Third Generation 3G

- Objective is to provide fairly high-speed wireless communication to support multimedia, data, and video in addition to voice
- Share the following design features:
  - Bandwidth
    - All 3G systems is to limit channel usage to 5 MHz
    - Receivers ability to resolve multipath when compared to narrower bandwidths
    - Available spectrum is limited by competing needs

4G/5G Cellular

Third Generation 3G

- Date rate
  - Target data rates are 144 (moving users) and 384 kbps (slow or stationary users)
- Multirate
  - Multiple fixed-datarate logical channels to a given user.
  - Traffic on each logical channel can be switched independently through the wireless and fixed networks to different destinations
  - Multiple simultaneous applications from a given user
  - Can efficiently use the available capacity for each service

4G/5G Cellular

Fourth Generation

4G

- Ultra-broadband Internet access for a variety of mobile devices, including laptops, smartphones, and tablets
- 4G networks support Mobile web access and high-bandwidth applications such as high definition mobile TV, mobile video conferencing, and gaming services
- Designed to maximize bandwidth and throughput while also maximizing spectral efficiency

4G/5G Cellular

Fourth Generation 4G

- Have the following characteristics:
  - Based on an all-IP packet switched network, uses IP Protocol for voice, video and data.
  - Support peak data rates upto 100 Mbps (high mobility) and upto 1 Gbps (low-mobility)
  - Dynamically share and use the network resources to support more simultaneous users per cell
  - Support smooth handovers across heterogeneous networks
  - Support high QoS for next-generation multimedia applications

4G/5G Cellular

Fourth Generation 4G

- In contrast to earlier generations
  - 4G systems do not support traditional circuit-switched telephony service, providing only IP telephony services.

4G/5G Cellular

Fifth Generation 5G

- By 2020 the huge amounts of data traffic generated by tablets and smartphones
- Augmented by an equally huge amount of traffic from the *Internet of Things (*which includes shoes, watches, appliances, cars, thermostats, door locks, and much more)

4G/5G Cellular

Fifth Generation 5G

- Focus will be on:
  - Mostly the speed is improved in 4G
  - Building more intelligence into the network, making the network more smarter
  - Meeting service quality demands by dynamic use of priorities
  - Adaptive network reconfiguration
  - Other network management techniques
    - NFV, AI, Network slicing, SDN

Cloud Computing

- First became available in the early 2000s
- Particularly targeted at large enterprises
- Has spread to small and medium-size businesses and recently to consumers
- Apple's iCloud was launched in 2012 and had 20 million users within a week of the launch
- Evernote launched in 2008 and approached 100 million users in less than six years
- In 2014 Google announced that Google Drive had almost a quarter of a billion active users

Cloud Computing Concepts

- Cloud computing, cloud networking, and cloud storage are important concepts
- Even all IT operations moved to an Internet-connected infrastructure known as enterprise Cloud Computing.
- Users of PC and mobile devices use this service to backup data, sync devices, and share, using personal cloud services
- The National Institute of Standards and Technology (NIST) defines the essential characteristics of cloud computing as:
  - Broad network access
  - Rapid elasticity
  - Measured service
  - On-demand self service
  - Resource pooling

Cloud Computing Concepts

- Broad network access
  - Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous platforms (for example, mobile phones, laptops, and PDAs) and other traditional or cloud-based software services.
- Rapid elasticity
  - Cloud computing enables you to expand and reduce resources according to your specific service requirement. For example, you may need a large number of server resources for the duration of a specific task.
  - You can then release these resources upon completion of the task.

Cloud Computing Concepts

- Measured service
  - Cloud systems automatically control and optimize resource use by leveraging a metering capability
    - for example, storage, processing, bandwidth, and active user accounts.
  - Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer of the utilized service.

Cloud Computing Concepts

- On-demand self service
  - A consumer can unilaterally provision computing capabilities
    - such as server time and network storage, as needed automatically without requiring human interaction with each service provider.

Cloud Computing Concepts

- Resource pooling
  - The provider's computing resources are pooled to serve multiple consumers using a multitenant model
    - resources dynamically assigned and reassigned according to consumer demand.
    - Examples of resources include storage, processing, memory, network bandwidth, and virtual machines.
  - There is a degree of location independence
    - The customer has no control or knowledge over the exact location of the provided resources, but specify location at a higher level of abstraction (for example, country, state, or data center).
  - Even private clouds tend to pool resources between different parts of the same organization.

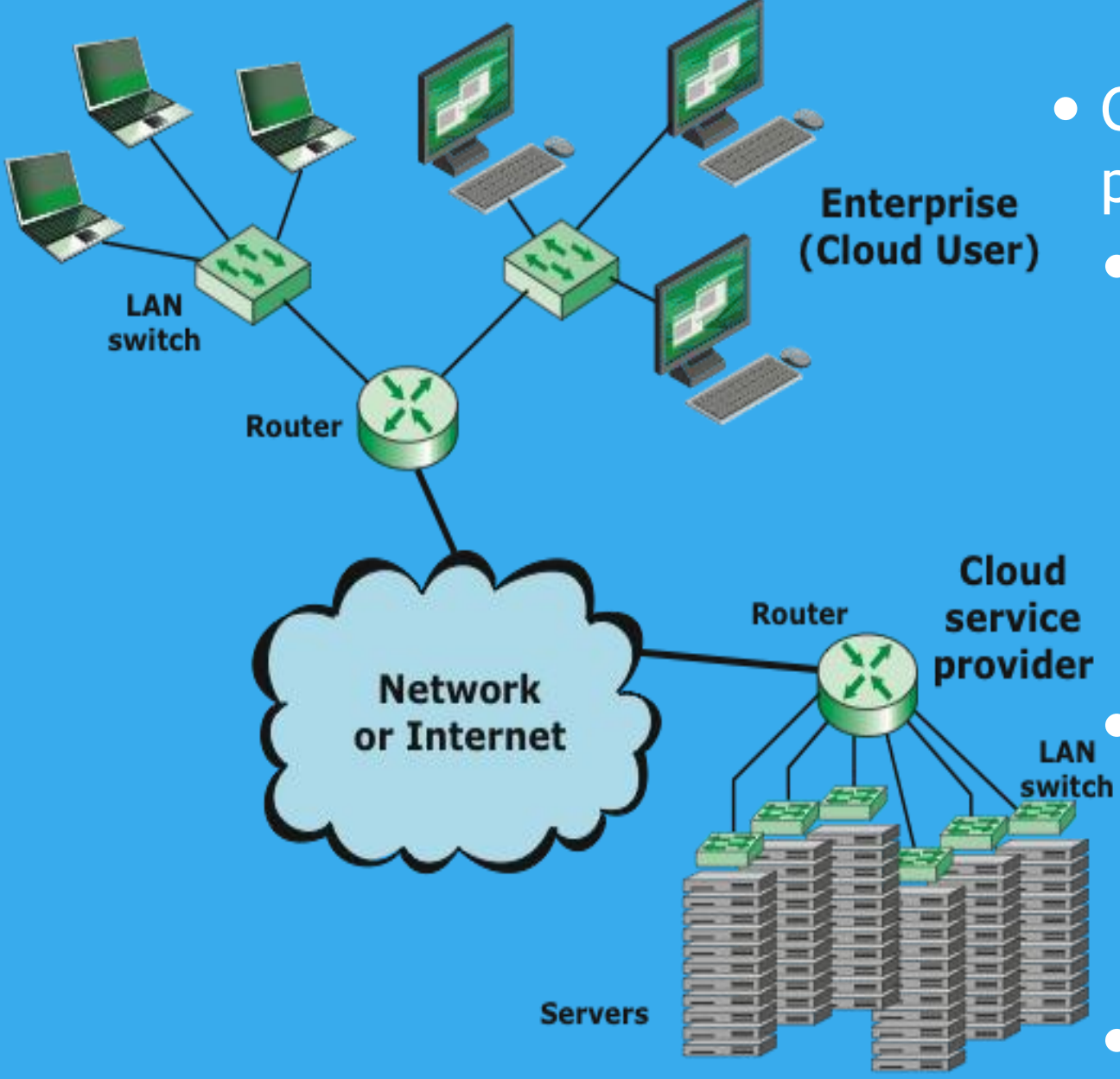


Figure 1.7 Cloud Computing Context

- Cloud service providers:
  - Maintains a massive collection servers, called as blade servers
  - Variety of network management
  - Security tools

Benefits of Cloud Computing

- Cloud computing provides:
  - Economies of scale
  - Professional network management
  - Professional security management
- Another big advantage of using cloud computing to store your data and share it with others is that the cloud provider takes care of security
  - Unfortunately the customer isn't always protected
  - There have been a number of security failures among cloud providers
    - Evernote told all of its users to reset their passwords after an intrusion was discovered.

Cloud Networking

- Refers to the networks and network management functionality that must be in place to enable cloud computing
  - Many cloud computing solutions rely on the Internet, but that is only a piece of the networking infrastructure
- One example is the provisioning high-performance/high-reliability networking between the provider and subscriber
  - In this case, some or all of the traffic between an enterprise and the cloud bypasses the Internet and uses dedicated private network facilities owned or leased by the cloud service provider

Cloud Networking

- More generally, cloud networking refers to the collection of network capabilities required to access a cloud, including:
  - Making use of specialized services over the Internet
  - Linking enterprise data centers to a cloud
  - Using firewalls and other network security devices at critical points to enforce access security policies

Cloud Storage

- Can be thought of as a subset of cloud computing
- Consists of database storage and database applications hosted remotely on cloud servers
- Enables small businesses and individual users to take advantage of data storage that scales with their needs and to take advantage of a variety of database applications without having to buy, maintain, and manage the storage assets

Internet of Things

- Term that refers to the expanding interconnection of smart devices, ranging from appliances to tiny sensors
- A dominant theme is the embedding of short-range mobile transceivers into a wide array of gadgets and everyday items, enabling new forms of communication between people and things and between things themselves
- Primarily driven by deeply embedded devices
- These devices are low-bandwidth, low-repetition data-capture and low-bandwidth data-usage appliances that communicate with each other and provide data via user interfaces

Evolution

- With reference to the end systems supported, the Internet has gone through roughly four generations of deployment culminating in the IoT:

Information technology (IT)

- PCs, servers, routers, firewalls, and so on, bought as IT devices by enterprise IT people, primarily using wired connectivity

Operation technology (OT)

- Machines/appliances with embedded IT built by non-IT companies, such as medical machinery, SCADA, process control, and kiosks, bought as appliances by enterprise OT people and primarily using wired connectivity

Personal technology

- Smartphones, tablets, and ebook readers bought as IT devices by consumers exclusively using wireless connectivity and often multiple forms of wireless connectivity

Sensor/actuator technology

- Single-purpose devices bought by consumers, IT, and OT people exclusively using wireless connectivity, generally of a single form, as part of larger systems

Layers of the Internet of Things

- Sensors and actuators
  - These are the “things”
  - Sensors observe their environment and report back quantitative measurements
  - Actuators operate on their environment
- Connectivity
  - A device may connect via either a wireless or wired link into a network to send collected data to the appropriate data center (sensor) or receive operational commands from a controller site (actuator)

Layers of the Internet of Things

- Capacity
  - The network supporting the devices must be able to handle a potentially huge flow of data
- Storage
  - There needs to be a large storage facility to store and maintain backups of all the collected data
- Data analytics
  - For large collections of devices, “big data” is generated, requiring a data analytics capability to process the data flow

Network Convergence

- Refers to the merger of previously distinct telephony and information technologies markets
- You can think of this convergence in terms of a three-layer model of enterprise communications:
  - Application convergence
  - Enterprise services
  - Infrastructure

Network Convergence

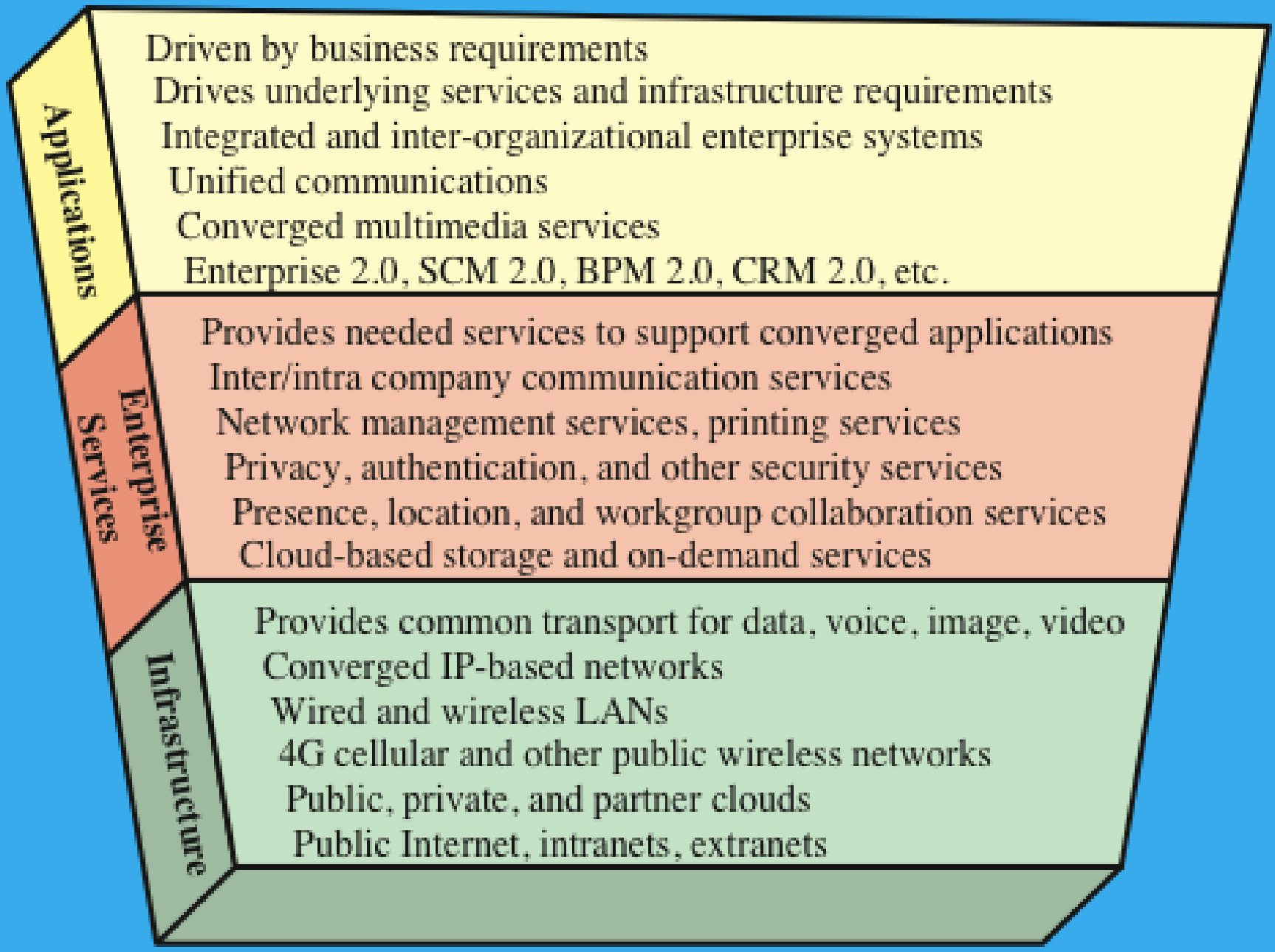
- Application convergence
  - These are seen by the end users of a business
  - Convergence integrates communications applications (Voice calling, voice mail, email and instant messaging) with business applications (workgroup collaboration, customer relationship management, back-office management)

Network Convergence

- Enterprise services
  - The manager deals with the information network in terms of the services that must be available to ensure that users can take full advantage of the applications that they use
    - Appropriate privacy mechanisms and authentication services are in place
    - Track user locations to support remote print services

Network Convergence

- Infrastructure
  - The network and communications infrastructure consists of the communication links, LANs, WANs, and Internet connections available to the enterprise
  - A key aspect of convergence at this level is the ability to carry voice, image, text, data and video over networks that were originally designed to carry data traffic.



Business driven convergence

Unified Communications (UC)

- Focuses on the integration of real-time communication services to optimize business processes
- IP is the cornerstone on which UC systems are built
- Key elements of UC include:
  - UC systems typically provide a unified user interface and consistent user experience across multiple devices and media
  - UC merges real-time communications services with non-real-time services and business process applications

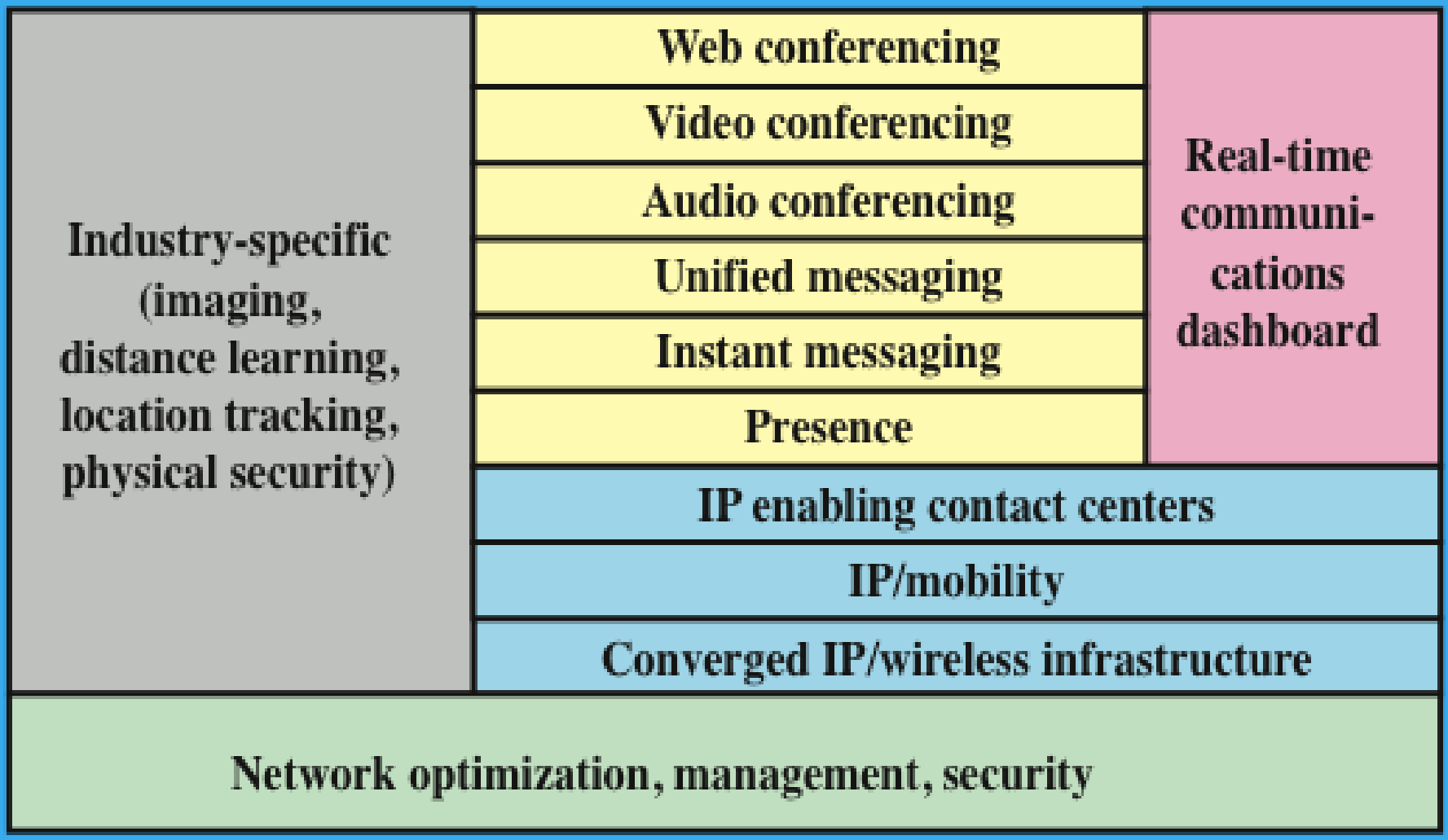


Figure 1.9 Elements of a Unified Communications Architecture

Unified Communications (UC)

- Real-time communications (RTC) dashboard
  - It provides UC users with a unified or consistent user interface across communication devices such as cell phone, desktop system, or office telephone system.
  - It provides access to real-time communication services such as instant messaging, audio and video conferencing;
  - Also provide access to non-real-time services such as unified messaging (e-mail, voice mail, fax, and SMS)
  - It includes presence information about co-workers and partners so that users to communicate or join a collaborative communication session.
  - They have become necessities in organizations that require high levels of communication and collaboration to support business processes.

Unified Communications (UC)

- Instant messaging (IM)
  - Real-time text-based messaging between two or more participants.
  - IM is distinct from online chat in that IM clients use contact (or buddy) lists to facilitate connections between known users, whereas online chat can include text-based exchanges between anonymous users.

Unified Communications (UC)

- Video teleconferencing (VTC)
  - Videoconferencing allows users in two or more locations to interact simultaneously via two-way video and audio transmission.
  - UC systems enable users to participate in video conferences via desktop computers, smartphones, and mobile devices.

Unified Communications (UC)

- Presence
  - The capability to determine, in real time, where someone is, how that person prefers to be reached, and even what the person is currently doing.
  - Presence information shows the individual's availability state before co-workers attempt to contact them person.
    - whether co-workers are currently on office or mobile phones, involved in a video call or in a meeting, or out of the office for lunch or vacation.
    - A co-worker's geographic location is becoming more common as an element in presence information for a number of business reasons, including the capability to quickly respond to customer emergencies.
    - Business has embraced presence information because it facilitates more efficient and effective communication.

Unified Communications (UC)

- IP enabling contact centers
  - Refers to the use of IP-based unified communications to enhance customer contact center functionality and performance.
  - The UC infrastructure makes use of presence technology to enable customers and internal enterprise employees to be quickly connected to the required expert or support person.
  - This technology supports mobility, so that call center personnel need not be located at a particular office or place.
  - This infrastructure enables the call center employee to quickly access other employees and information assets, including data, video, image, and audio.

Unified Communications (UC)

- IP/mobility:
  - Refers to the delivery of information to and collection of information from enterprise personnel who are usually mobile, using an IP network infrastructure.
  - In a typical enterprise, upward of 30 percent of employees use some form of weekly remote access technology in the performance of their jobs.

Unified Communications (UC)

- Converged IP/wireless infrastructure
  - A unified networking and communications-based IP packet transfer to support voice, data, and video transmission and
    - extended to include local- and wide-area wireless communications.
  - UC-enabled mobile devices are able to switch between Wi-Fi and cellular systems in the middle of a communication session.
  - Both handoffs (home Wi-Fi to cellular and cellular to office Wi-Fi) would take place seamlessly and transparently without dropping the call.

Unified Communications (UC)

- Major benefits
  - Personal productivity gains
    - Calls from VIP contacts can be routed simultaneously to all of a UC user's phone devices(office and home phone etc)
    - Faster respensiveness
  - Workgroup performance gains
    - Speed identification of an available individual with the right skills a work team needs
  - Enterprise-level process improvements
    - Redefining best practices for customer relationship management, Supply chain management etc

End of Chapter 1

Foundations of Modern Networking

SDN, NFV, QoE, IoT, and Cloud

By: William Stallings

Chapter 2

Requirements and Technology

Types of Network and Internet Traffic

- Elastic Traffic
  - Can adjust, over wide ranges, to changes in delay and throughput across an internet and still meet the needs of its applications
    - "internet" refers to a large network made up of a number of smaller networks
    - "Internet" refers to a worldwide inter-network based on TCP/IP that interconnects thousands of public and private networks.

Types of Network and Internet Traffic

- Elastic Traffic
  - This is the traditional type of traffic supported on TCP/IP based internets
  - This is the type of traffic for which internets were designed
  - Applications that generate such traffic typically use Transmission Control Protocol (TCP) or User Datagram Protocol (UDP) as a transport protocol

Types of Network and Internet Traffic

- Elastic Traffic
  - In UDP, the application will use the capacity up to the rate that the application generates data.
  - In TCP, the application will use the capacity up to the maximum rate that the end-to-end receiver can accept data.
    - Traffic on individual connections adjusts to congestion by reducing the rate at which data are presented to the network.

Types of Network and Internet Traffic

- Elastic Traffic
  - Applications:
    - file transfer (File Transfer Protocol / Secure FTP [FTP/SFTP])
    - electronic mail (Simple Mail Transport Protocol [SMTP])
    - remote login (Telnet, Secure Shell [SSH])
    - network management (Simple Network Management Protocol [SNMP])
    - web access (Hypertext Transfer Protocol / HTTP Secure [HTTP/HTTPS]).

Types of Network and Internet Traffic

- There are differences among the requirements of these applications:
  - E-mail is generally insensitive to changes in delay.
  - When file transfer is done via user command, the user expects the delay to be proportional to the file size and so is sensitive to changes in throughput.
  - With network management, delay is generally not a serious concern.
    - The need for SNMP messages to get through with minimum delay increases with increased congestion.
  - Interactive applications, such as remote login and web access, are sensitive to delay.

Types of Network and Internet Traffic

- Elastic Traffic
  - It is not the per-packet delay
  - Real delays across the Internet suggest that wide variations in delay do not occur. Because of the congestion control mechanisms in TCP.
    - For an interactive Telnet-based application, the element may be a single keystroke or single line.
    - For web access, the element is a web page (as little as a few kilobytes or larger for an image-rich page).
    - For a scientific application, the element could be many megabytes of data.
      - For large elements, total elapsed time is dictated by the sliding-window performance of TCP

Types of Network and Internet Traffic

- Inelastic Traffic
  - Does not easily adapt to changes in delay and throughput across an internet
  - Examples of inelastic traffic include:
    - multimedia transmission such as voice and video,
    - high-volume interactive traffic

Types of Network and Internet Traffic

- Inelastic traffic requirements may include:
  - Throughput
    - A minimum throughput value may be required
  - Delay
    - Also called latency. Example, stock trading
  - Delay jitter
    - The magnitude of delay variation, called delay jitter, is a critical factor in real-time applications
    - Interarrival times between packets are not maintained at a fixed interval at the destination
    - Buffering is used
  - Packet loss
    - Real time applications that can sustain vary in packet loss rate

Table 2.1 Service Class Characteristics

Application Category	Service Class	Traffic Characteristics	Tolerance to		
			Loss	Delay	Jitter
Control	Network control	Variable-size packets, mostly inelastic short messages, but traffic can also burst (BGP)	Low	Low	Yes
	OA&M	Variable-size packets, elastic and inelastic flows	Low	Medium	Yes
Media-Oriented	Telephony	Fixed-size small packets, constant emission rate, inelastic and low-rate flows	Very low	Very low	Very low
	Real-time interactive	RTP/UDP streams, inelastic, mostly variable rate	Low	Very low	Low
	Multimedia conferencing	Variable-size packets, constant transmit interval, rate adaptive, reacts to loss	Low-medium	Very low	Low
	Broadcast video	Constant and variable rate, inelastic, non-bursty flows	Very low	Medium	Low
	Multimedia Streaming	Variable-size packets, elastic with variable rate	Low-medium	Medium	Yes
Data	Low-latency data	Variable rate, bursty short-lived elastic flows	Low	Low-medium	Yes
	High-throughput data	Variable rate, bursty long-lived elastic flows	Low	Medium-high	Yes
	Low-priority data	Non-real-time and elastic	High	High	Yes
Best effort	Standard	A bit of everything	Not specified		

BGP = Border Gateway Protocol
OA&M = operations, administration, and management
RTP = Real-Time Transport Protocol
UDP = User Datagram Protocol

Table 2.2 QoS Requirements by Application Class

Voice	One-way latency ≤ 150 ms. One-way peak-to-peak jitter ≤ 30 ms. Per-hop peak-to-peak jitter ≤ 10 ms. Packet loss ≤ 1 percent.
Broadcast video	Packet loss ≤ 0.1 percent.
Real-time interactive video	One-way latency ≤ 200 ms. One-way peak-to-peak jitter ≤ 50 ms. Per-hop peak-to-peak jitter ≤ 10 ms. Packet loss ≤ 0.1 percent
Multimedia conferencing	One-way latency ≤ 200 ms. Packet loss ≤ 1 percent.
Multimedia streaming	One-way latency ≤ 400 ms. Packet loss ≤ 1 percent.

Real-Time Traffic Characteristics

- An example of inelastic traffic
- Are concerned with timing issues as well as packet loss
  - In elastic applications such e-mail, file transfer
    - Mechanisms are there to make sure no data are lost, corrupted and disordered during the transit
- In most cases there is a requirement that data be delivered at a constant rate equal to the sending rate
- In other cases a deadline is associated with each block of data, such that the data are not usable after the deadline has expired

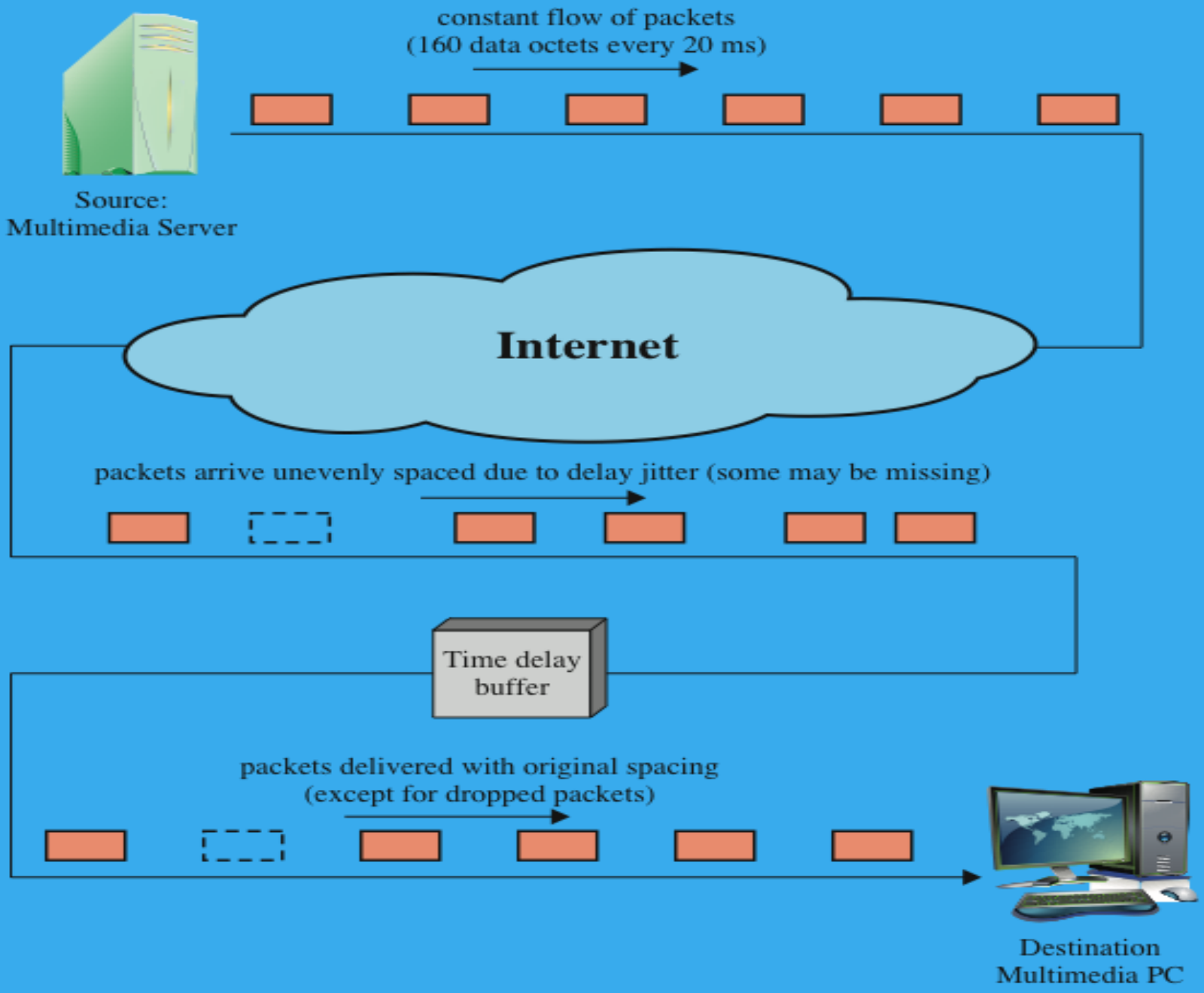
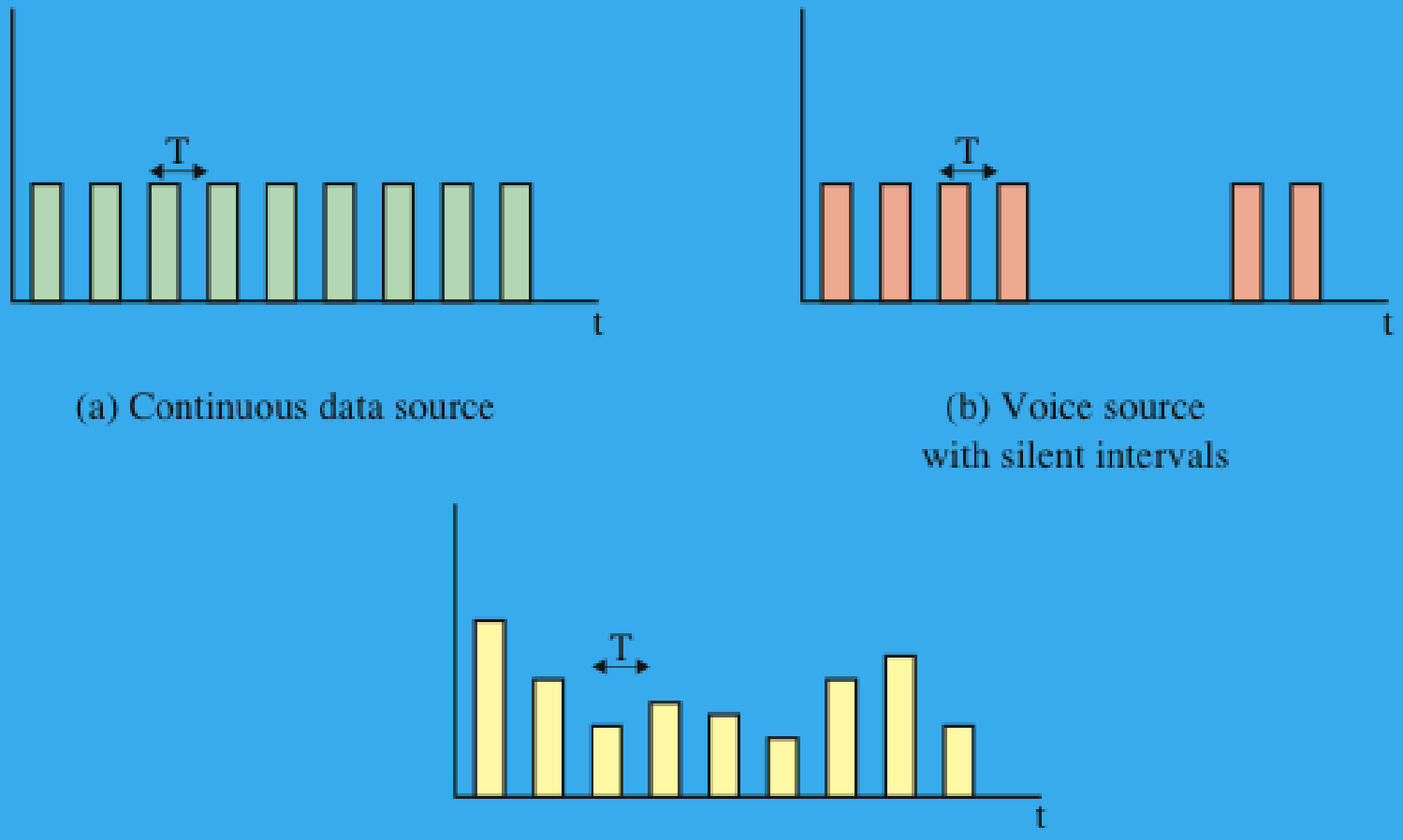


Figure 2.1 Real-Time Traffic



(a) Continuous data source

(b) Voice source with silent intervals

(c) Compressed video source

Figure 2.2 Real-Time Packet Transmission

Real-Time Traffic Characteristics

- Continuous data source:
  - Fixed-size packets are generated at fixed intervals.
  - This characterizes applications that are constantly generating data
    - Have few redundancies, and that are too important to compress in a lossy way.
    - Accuracy and reliability matters not bandwidth saving
  - Examples are air traffic control radar and real-time simulations.

Real-Time Traffic Characteristics

- On/off source:
  - The source alternates between periods when fixed-size packets are generated at fixed intervals and periods of inactivity.
  - A voice source, such as in telephony or audio conferencing, fits this profile.
- Variable packet size:
  - The source generates variable-length packets at uniform intervals.
  - An example is digitized video in which different frames may experience different compression ratios for the same output quality level.

Big Data

- Refers to everything that enables an organization to create, manipulate, and manage very large data sets and the facilities in which these are stored
  - Terabytes, petabytes, exabytes etc
  - Government regulations and court cases that have caused organizations to stockpile large masses of data, including documents, e-mail messages, voice-mail messages, etc

Big Data

- Infrastructure
  - Rational database management systems (RDBMS)
  - Network-attached storage (NAS)
  - Storage area networks (SANs)
  - Data warehouses (DWs)
  - Business intelligence (BI) analytics

Big Data

- Traditional data warehouse and BI analytics systems tend to be highly centralized within an enterprise infrastructure.
  - Central data repository with a RDBMS
  - High-performance storage
  - Analytics software such OLAP tools for mining and visualizing data

- other important network devices, including firewalls, IDS/IPS, LAN switches, and routers.

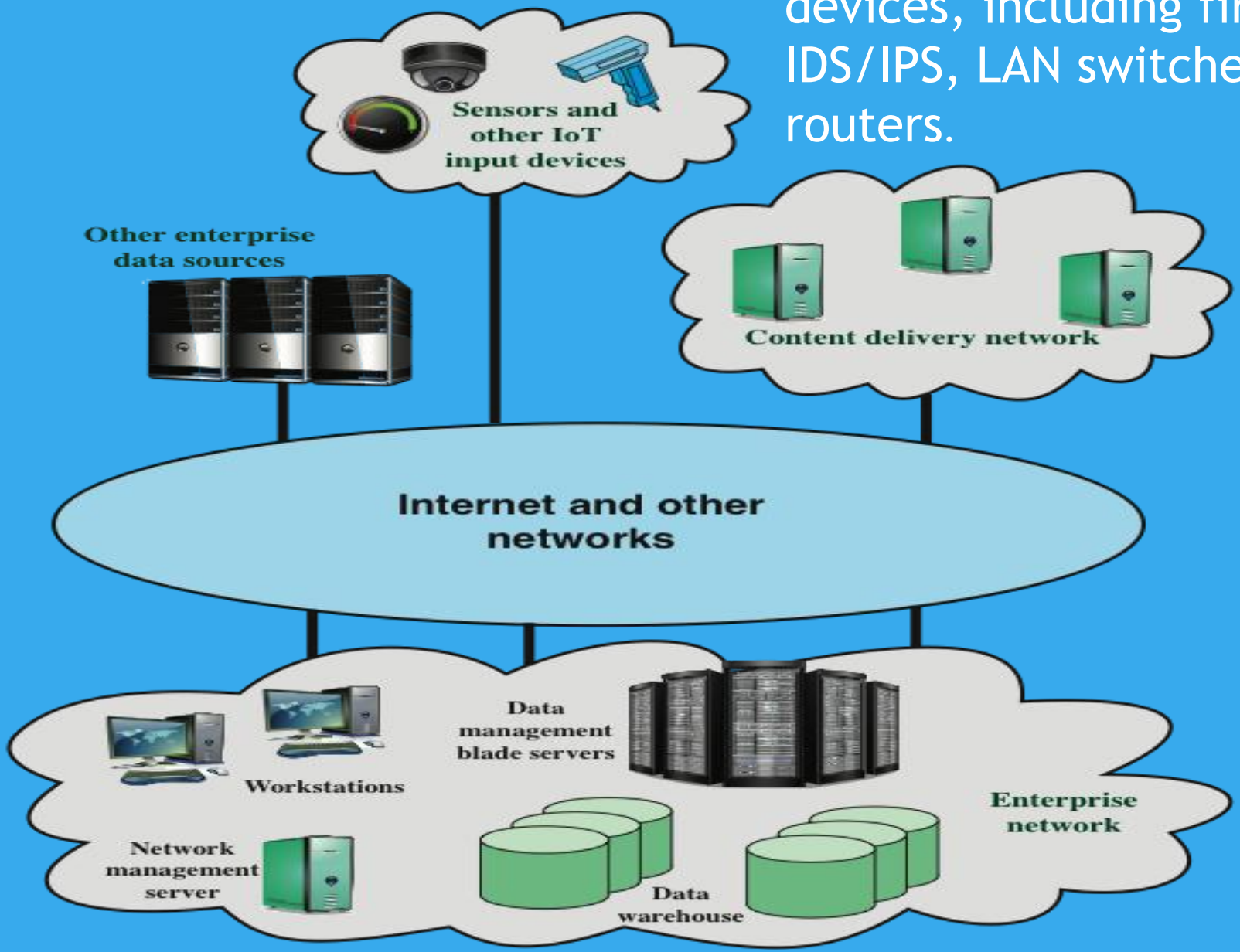


Figure 2.3 A Big Data Networking Ecosystem

Key elements within the enterprise :

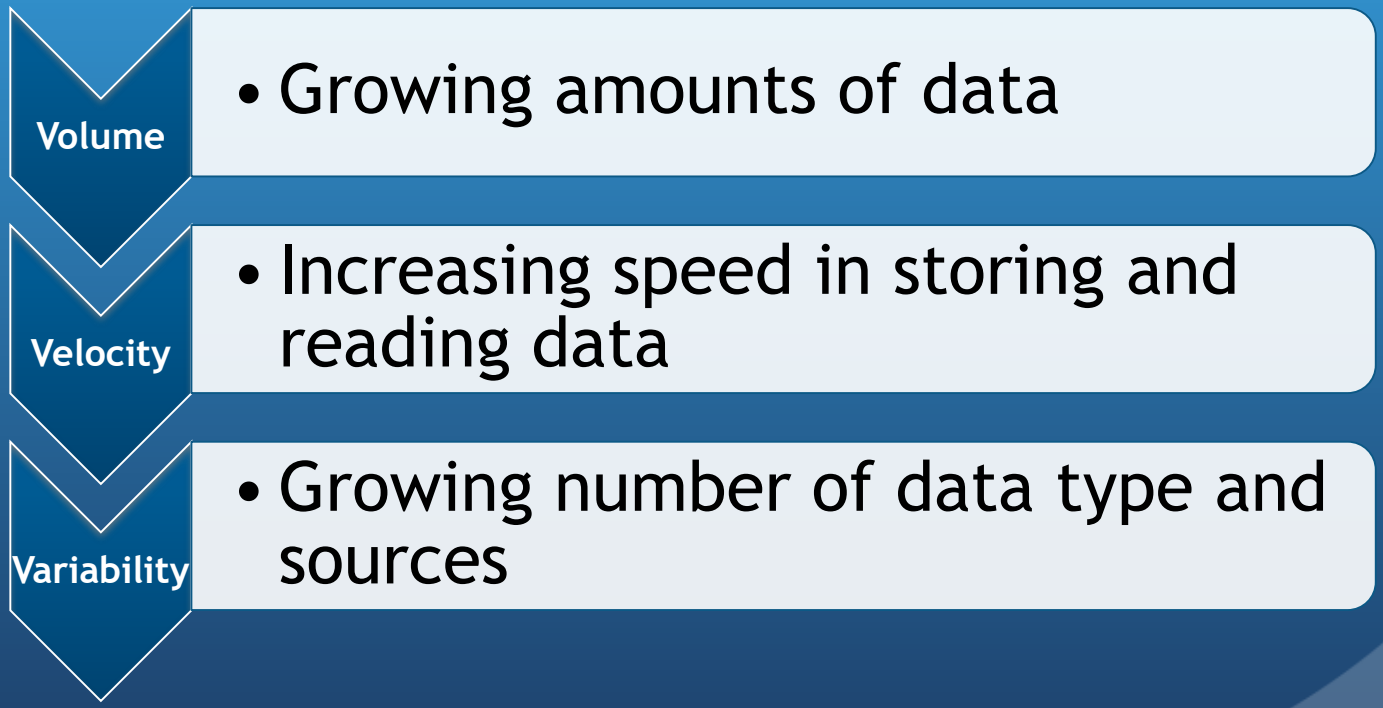
- Data warehouse:
  - The DW holds integrated data from multiple data sources, used for reporting and data analysis.

- Data management servers:
  - Large banks of servers serve multiple functions with respect to big data.
  - The servers run data analysis applications, such as data integration tools and analytics tools.
- Workstations / data processing systems:
  - use of big data applications and in the generation of input to big data warehouses.

- Network management server:
  - One or more servers responsible for network management, control, and monitoring.

Big Data

- The impact of big data on an enterprise's networking infrastructure is driven by the so-called three V's:



Areas of Concern:

Network capacity

- Running big data analytics requires a lot of capacity on its own;
  - The issue is magnified when big data and day-to-day application traffic are combined over an enterprise network

Latency

- The real or near-real-time nature of big data demands a network architecture with consistent low latency to achieve optimal performance
- Example, fraud detection in bank transactions

Storage capacity

- Massive amounts of highly scalable storage are required,
  - These resources must be flexible enough to handle many different data formats and traffic loads

Processing

- Big data can add significant pressure on computational memory and storage systems, which, if not properly addressed, can negatively impact operational efficiency

Secure data access

- Big data projects combine sensitive information from many sources like customer transactions, GPS coordinated, video streams, and so on, which must be protected from unauthorized access.

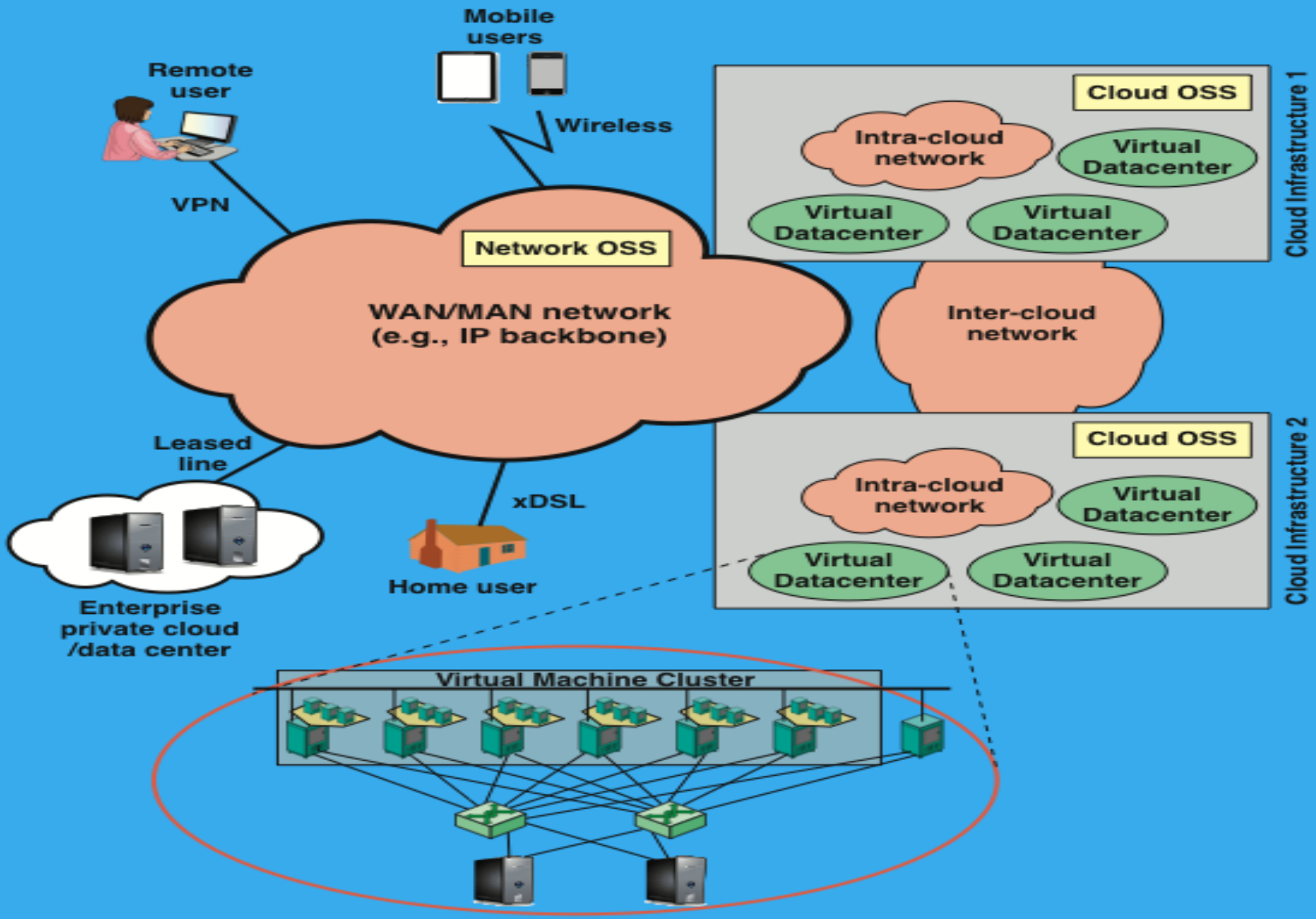


Figure 2.4 Cloud Network Model

Cloud Computing

- Intracloud, intercloud, core, together with the OSS components, are the foundation of cloud services composition and delivery
- Intracloud connects database servers, storage arrays, and other servers such as firewalls, load balancers, IDS etc.
- Intracloud network includes a number of LANs interconnected with IP routers.
  - Database servers are organized as a cluster of Virtual machines
- Intercloud networks interconnect cloud infrastructures together

Cloud Computing

Two categories of operations support system (OSS):

- Network OSS:
  - The processes supported by a network OSS
    - service management and maintenance of the network inventory,
    - configuration of particular network components,
    - fault management.
- Cloud OSS:
  - It is the system dedicated to providers of cloud computing services.
  - Cloud OSS supports processes for the maintenance, monitoring, and configuration of cloud resources.

Cloud Computing

- Functional requirements for this network capability:
  - Scalability
    - Dedicated to providers of telecommunication services
    - Networks must be able to scale easily
      - to meet the demands of moving from current cloud infrastructures of hundreds or a few thousand servers to networks of tens or even hundreds of thousands of servers

Cloud Computing

- Functional requirements for this network capability:
  - Performance
    - Traffic in both big data installations and cloud provider networks is unpredictable and quite variable
    - The current three-tier topology (access, aggregation, and core) used in data centers is not well adapted to provide these requirements.
    - A more flexible and dynamic control of data flows, in addition to virtualization of network devices

Cloud Computing

- Functional requirements for this network capability:
  - Agility and flexibility
    - The cloud-based data center needs to be able to respond and manage the highly dynamic nature of cloud resource utilization

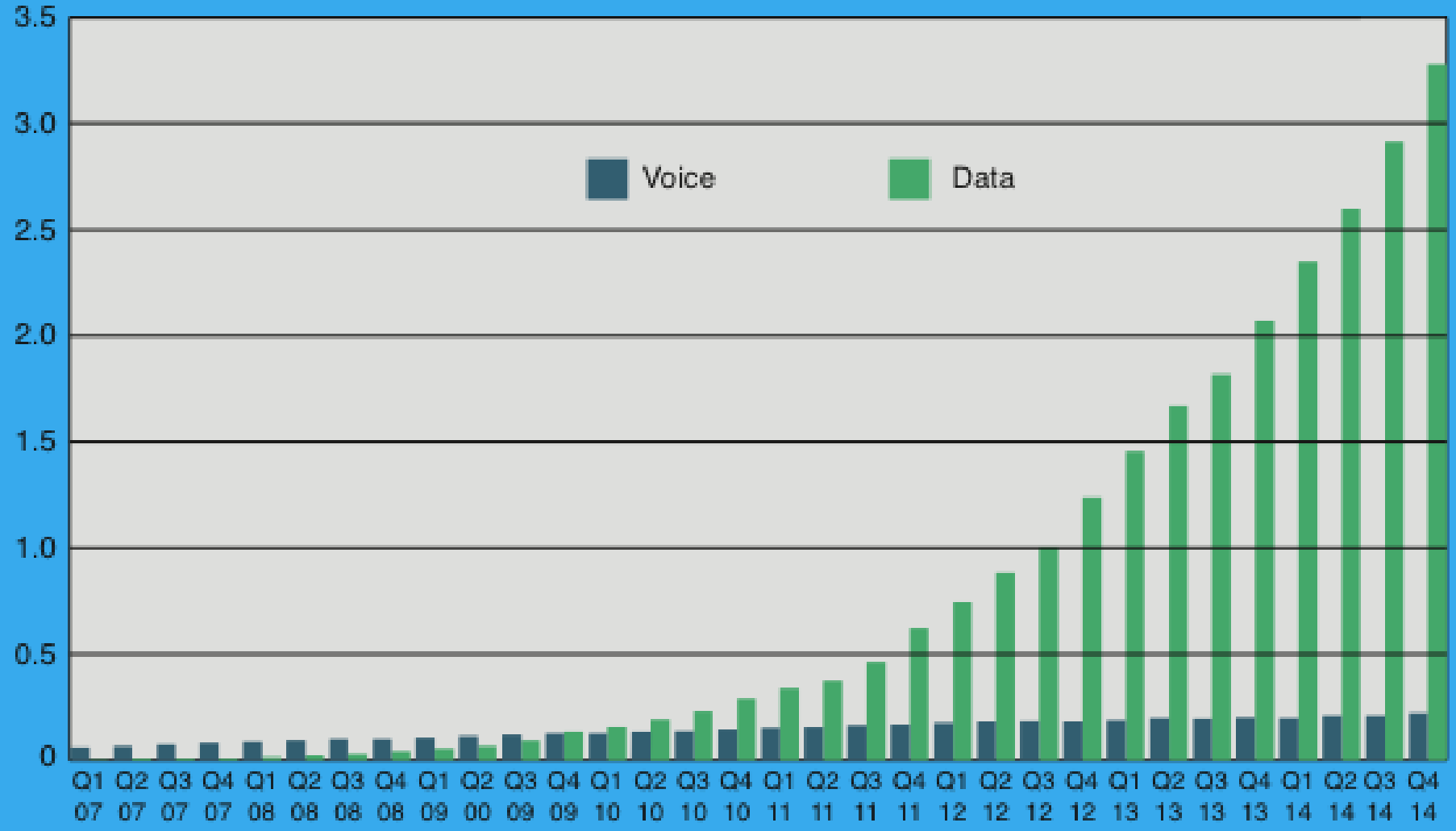


Figure 2.5 World Total Monthly Mobile Voice and Data Traffic (exabytes/month) [AKAM15]

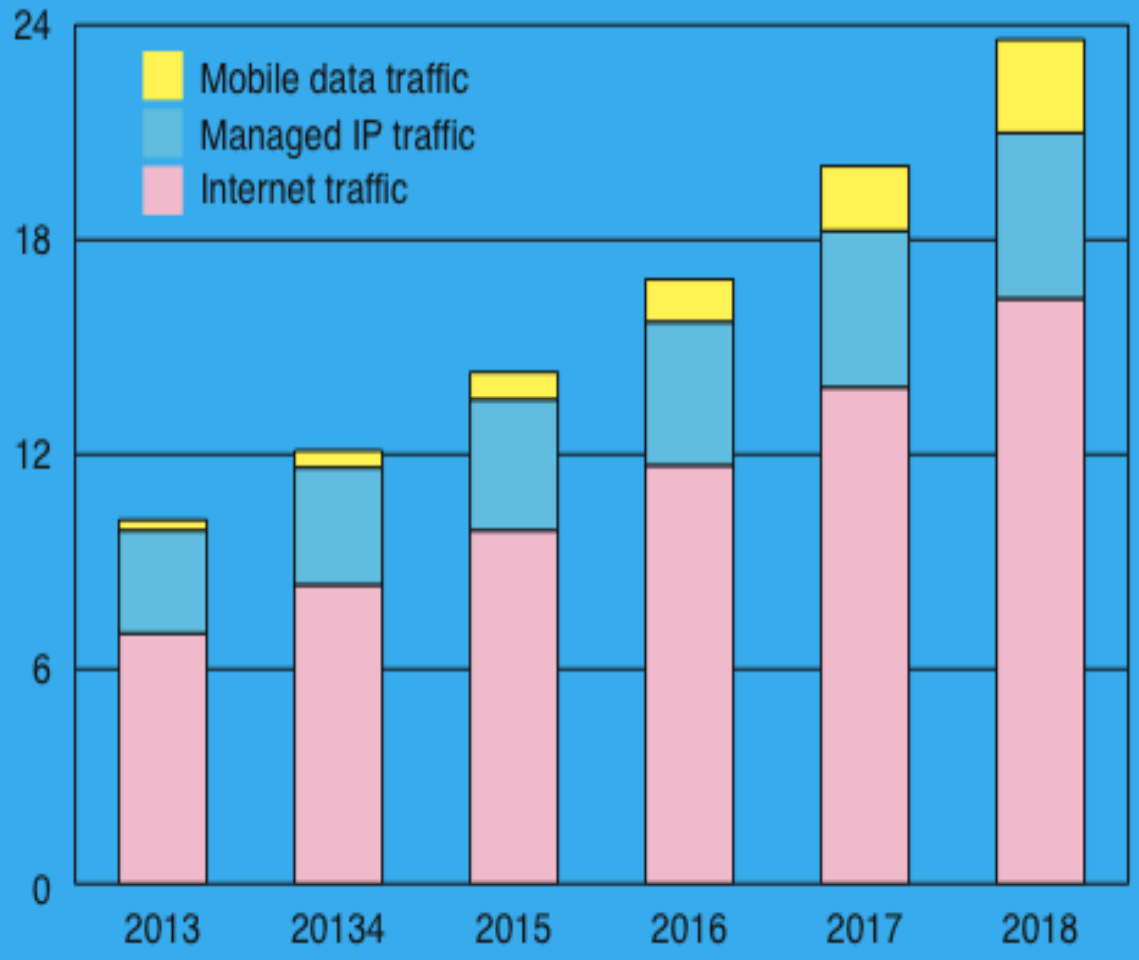


Figure 2.6 Forecast Monthly Enterprise IP Traffic (exabytes/month) [CISC14]

- **Mobile data traffic:** All enterprise traffic that crosses a mobile access point
- **Managed IP traffic:** All enterprise traffic that is transported over IP but remains within the corporate WAN
- **Internet traffic:**
- All enterprise traffic that crosses the public Internet

Quality of Service (QoS)

- The measurable end-to-end performance properties of a network service
  - Guaranteed in advance by a service level agreement (SLA) between a user and a service provider, so as to satisfy specific customer application requirements

Quality of Service (QoS)

Commonly specified properties include:

- Throughput: bytes or bits per second, for a given logical connection or traffic flow
- Delay
- Packet jitter
- Error rate
- Packet loss
- Priority
- Availability
- Security

Quality of Experience (QoE)

- A subjective measure of performance as reported by the user; relies on human opinion
- Is important particularly when dealing with multimedia applications and multimedia content delivery
- QoS processes by themselves are not sufficient in that they do not take into account the user's perception of network performance and service quality
- There are numerous ways the multimedia content could have been encoded, giving rise to differing perceived qualities.

Quality of Experience (QoE)

Categories of factors and features that can be included in QoE are:

- Perceptual
  - The quality of the sensory aspects of user experience
    - For video, examples, Sharpness, brightness, contrast, distortion etc.
    - For Audio, examples, clarity
- Psychological
  - Ease of use, joy of use, usefulness, perceived quality, satisfaction, annoyance etc.
- Interactive
  - Interaction between the user and the application or device
  - Examples, responsiveness, naturalness of interaction, communication efficiency and accessibility

Routing Characteristics

- The primary function of an internet is to accept packets from a source station and deliver them to a destination station
  - To accomplish this, a path or route through the network must be determined
  - Generally, more than one route is possible; therefore, a routing function must be performed

Routing Characteristics

- Selection of a route is generally based on some performance criterion
  - Simplest criterion is to choose the minimum-hop route (one that passes through the least number of nodes) through the network
  - A generalization of the minimum-hop criterion is least-cost routing;
    - a cost is associated with each link, and, for any pair of attached stations, the route through the network that accumulates the least cost is sought

- The shortest path (fewest hops) from node 1 to node 6 is 1-3-6 (Cost = 5 + 5 = 10), but the least-cost path is 1-4-5-6 (Cost = 1 + 1 + 2 = 4).

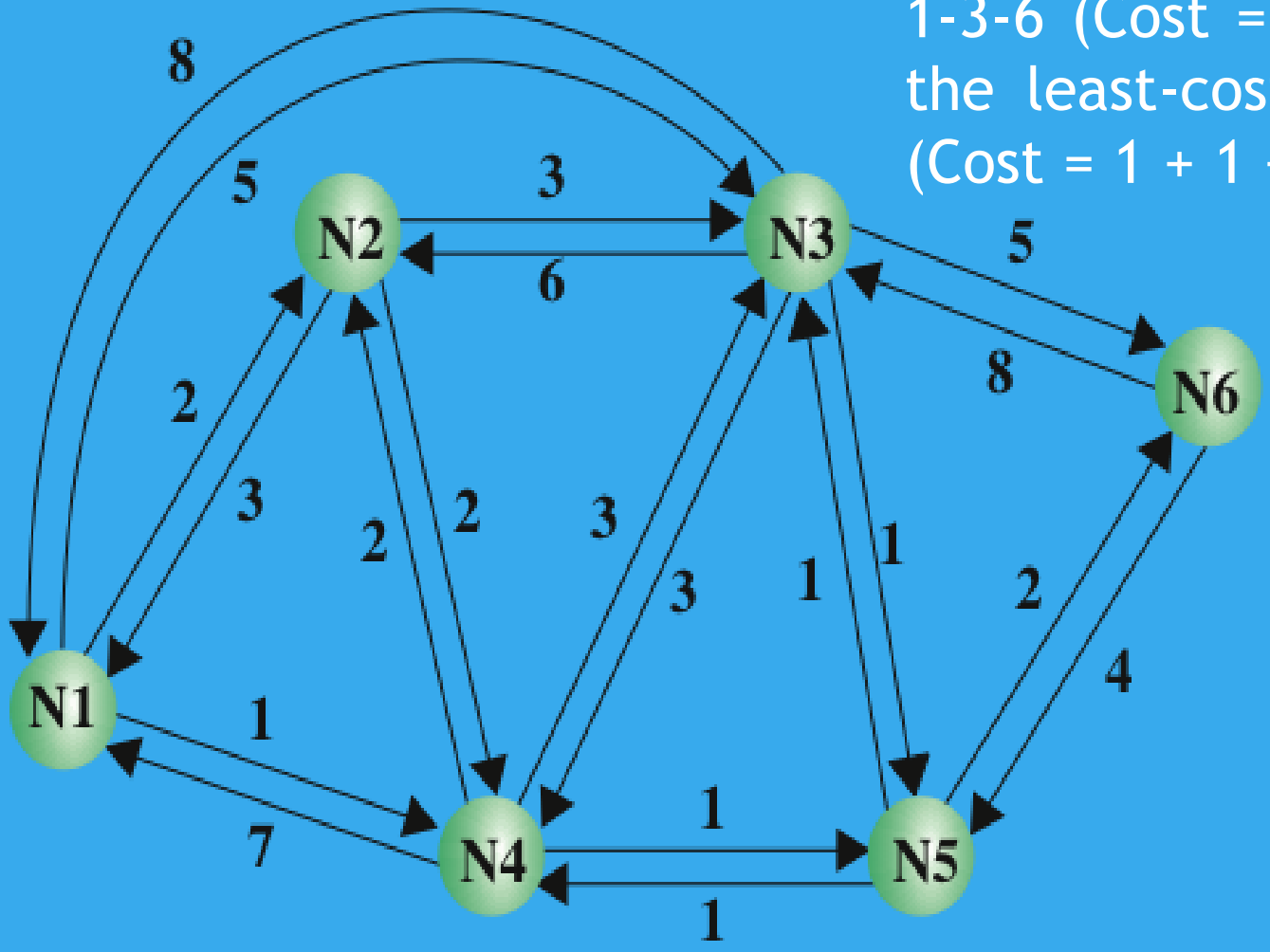


Figure 2.7 Example Network Architecture

CENTRAL FORWARDING TABLE

To Node \ From Node	1	2	3	4	5	6
1	—	1	5	2	4	5
2	2	—	5	2	4	5
3	4	3	—	5	3	5
4	4	4	5	—	4	5
5	4	4	5	5	—	5
6	4	4	5	5	6	—

Node 1 Table

Destination	Next Node
2	2
3	4
4	4
5	4
6	4

Node 2 Table

Destination	Next Node
1	1
3	3
4	4
5	4
6	4

Node 3 Table

Destination	Next Node
1	5
2	5
4	5
5	5
6	5

Node 4 Table

Destination	Next Node
1	2
2	2
3	5
5	5
6	5

Node 5 Table

Destination	Next Node
1	4
2	4
3	3
4	4
6	6

Node 6 Table

Destination	Next Node
----	5
2	5
3	5
4	5
5	5

Figure 2.8 Packet Forwarding Tables (using Figure 2.7)

- Additional information used to determine the forwarding decision, such as the source address, packet flow identifier, or security level of the packet:
  - Failure: When a node or link fails, it can no longer be used as part of a route.
  - Congestion: When a particular portion of the network is heavily congested, route packets around congestion area.
  - Topology change: The insertion of new links or nodes affects routing.
- For adaptive routing, the network state information must be exchanged among the nodes or between the nodes and a central controller.

Routing Protocols

- Routers are responsible for receiving and forwarding packets through the interconnected set of networks
- Each router makes routing decisions based on knowledge of the topology and traffic/delay conditions of the internet
- A degree of dynamic cooperation is needed among the routers
- The router must avoid portions of the network that have failed and should avoid portions of the network that are congested

Routing Protocols

- There are essentially two categories of routing protocols which are based on the concept of an autonomous system (AS)
- An AS exhibits the following characteristics:
  - An AS is a set of routers and networks managed by a single organization
  - An AS consists of a group of routers exchanging information via a common routing protocol
  - Except in times of failure, an AS is connected (in a graph-theoretic sense)
- An interior router protocol (IRP) is a shared routing protocol that passes routing information between routers within an AS

Routing Protocols

- The routers in one AS need at least a minimal level of information concerning networks outside the system that can be reached.
  - The protocol used to pass routing information between routers in different autonomous systems as an exterior router protocol (ERP) .
- You can expect that an ERP will need to pass less information than an IRP.

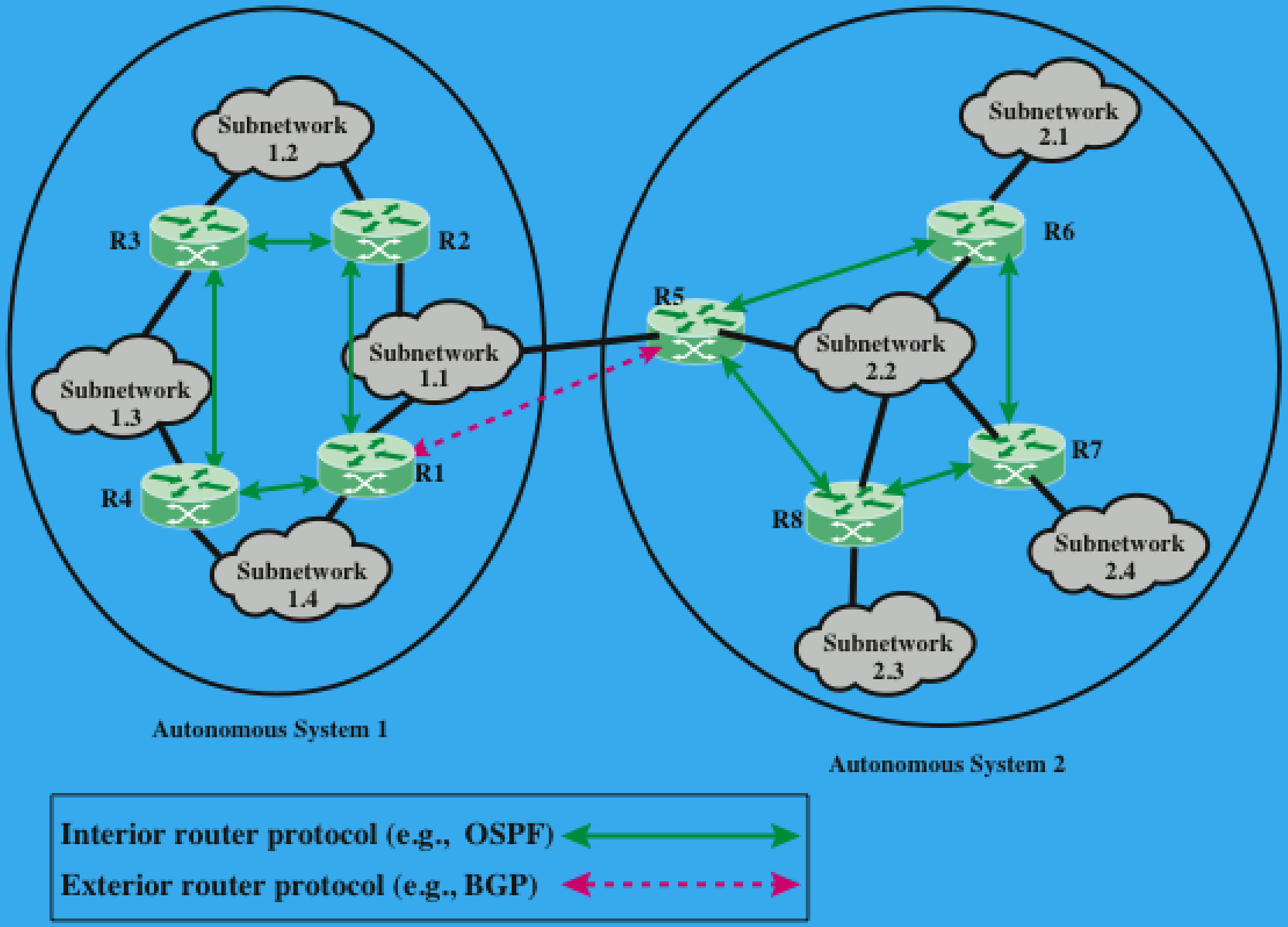


Figure 2.9 Use of Exterior and Interior Routing Protocols

The principal elements of a router

- Any given router has a number of I/O ports:
  - one or more to other routers, and zero or more to end systems.
    - On each port, packets arrive and depart.
    - Two buffers, or queues, at each port:
      - one to accept arriving packets, and
      - one to hold packets that are waiting to depart.
  - In practice, two fixed-size buffers associated with each port
  - You can think of each port having two variable-size buffers
    - subject to the constraint that the sum of all buffer sizes is a constant.

The principal elements of a router

- As packets arrive, they are stored in the input buffer of the corresponding port.
  - The router examines each incoming packet
    - makes a routing decision based on the forwarding tables, and then moves the packet to the appropriate output buffer.

The principal elements of a router

- Each output queue can be operated as a simple first-in, first-out (FIFO) queue.
  - More commonly, a more complex queuing discipline is used, to take into account the relative priority of the queued packets.
  - A set of routing policies may also influence the construction of the forwarding tables and how various packets are to be treated.
    - Policies may determine routing not just on the destination address but other factors, such as source address, packet size, and protocol of the payload.

The principal elements of a router

- A routing control function includes:
  - execution of routing protocols,
  - adaptive maintenance of the routing tables, and
  - supervising congestion control policies.

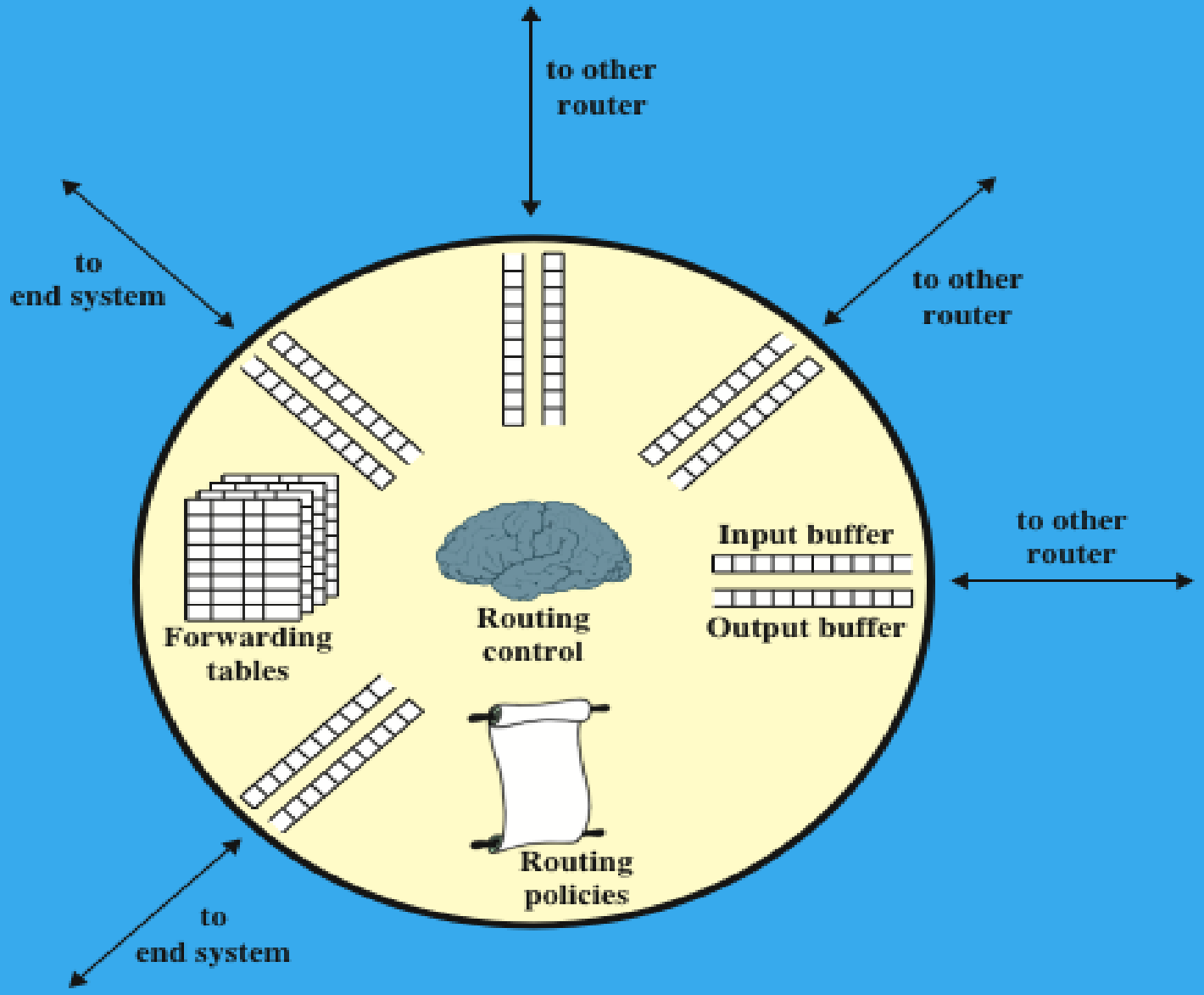


Figure 2.10 Elements of a Router

Congestion Control

If packets arrive too fast for a router to process them or faster than packets can be cleared from the outgoing buffers, eventually, no memory is available.

- The first such strategy is to discard any incoming packet for which there is no available buffer space.
- The alternative is to exercise some sort of flow control over its neighbors so that the traffic flow remains manageable.

Congestion Control

- Congestion at one point in the network can quickly propagate throughout a region or the entire network.
- You need to use flow control in such a way as to manage the traffic on the entire network.

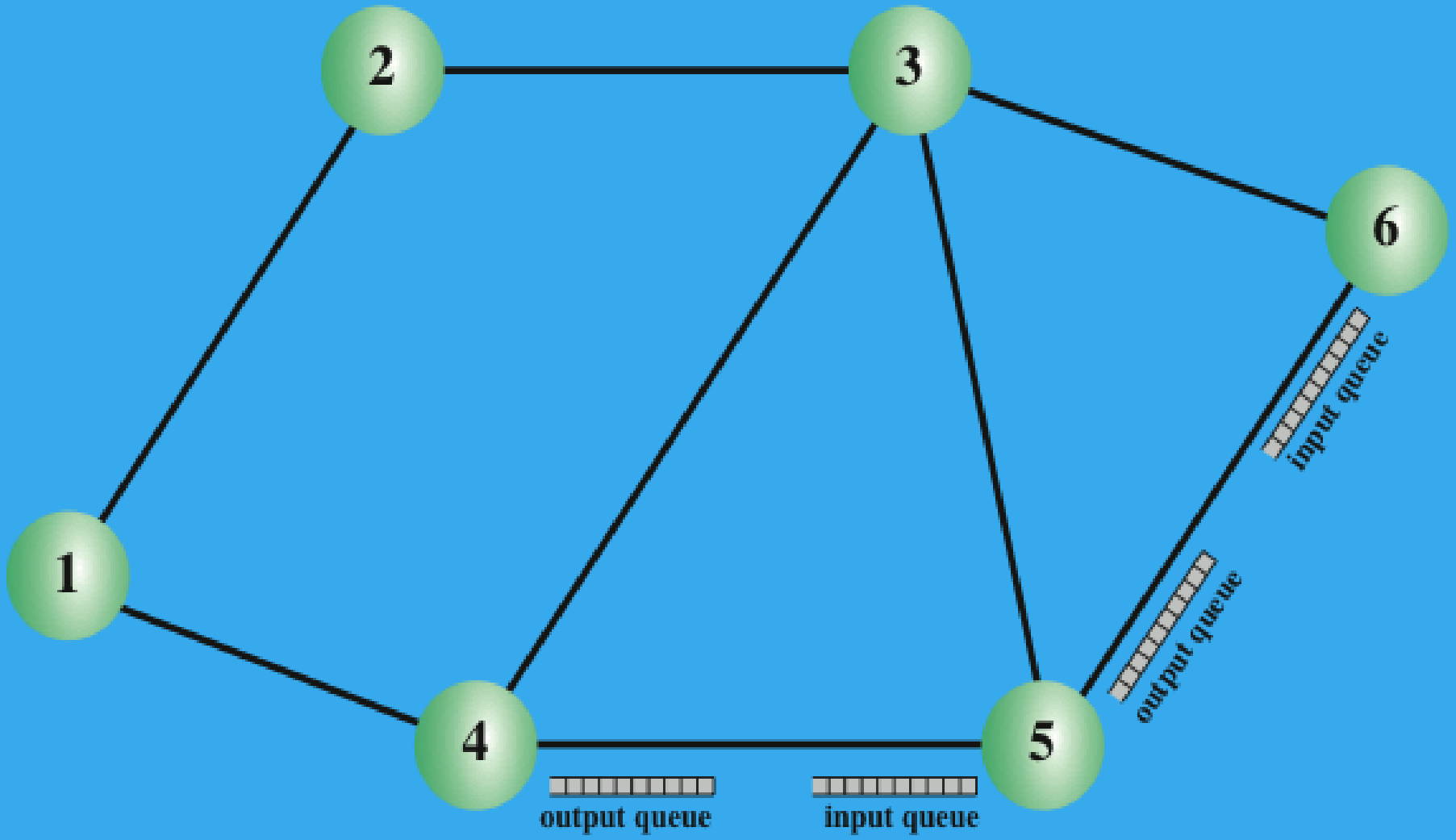


Figure: Interaction of Queues in a Data Network

Congestion Control

- The steady-state total throughput through the network as a function of the offered load,
  - both normalized to the maximum theoretical throughput of the network.

- In the ideal case, the throughput of the network increases up to the full capacity of the network; then normalized throughput remains at 1.0 at higher input loads.

- At negligible load, there is some small constant amount of delay (the propagation delay through source to destination plus processing delay at each node).

Congestion Control

- As the load on the network increases, queuing delays at each node are added.
  - Even if each source produced packets at fixed intervals, there will be fluctuation in the input rate at each individual network node.
    - When a burst of packets arrives at a node, it will take some time to clear the backlog. Thus, imposing packet bursts on downstream node.
  - And once a queue builds up at a node, even if the node can handle during a given time period, those packets have to wait their turn in the queue, and thus experience additional delay.

Congestion Control

- When the load exceeds the network capacity, delays increase without bound.
  - Suppose that each node in the network is equipped with buffers of infinite size and suppose that the input load exceeds network capacity.
  - Under ideal conditions, the rate of packets leaving the network is 1.0.
    - Because the rate of packets entering the network is greater than 1.0, internal queue sizes grow.
  - In the steady state, with input greater than output, these queue sizes grow without bound and therefore queuing delays grow without bound.

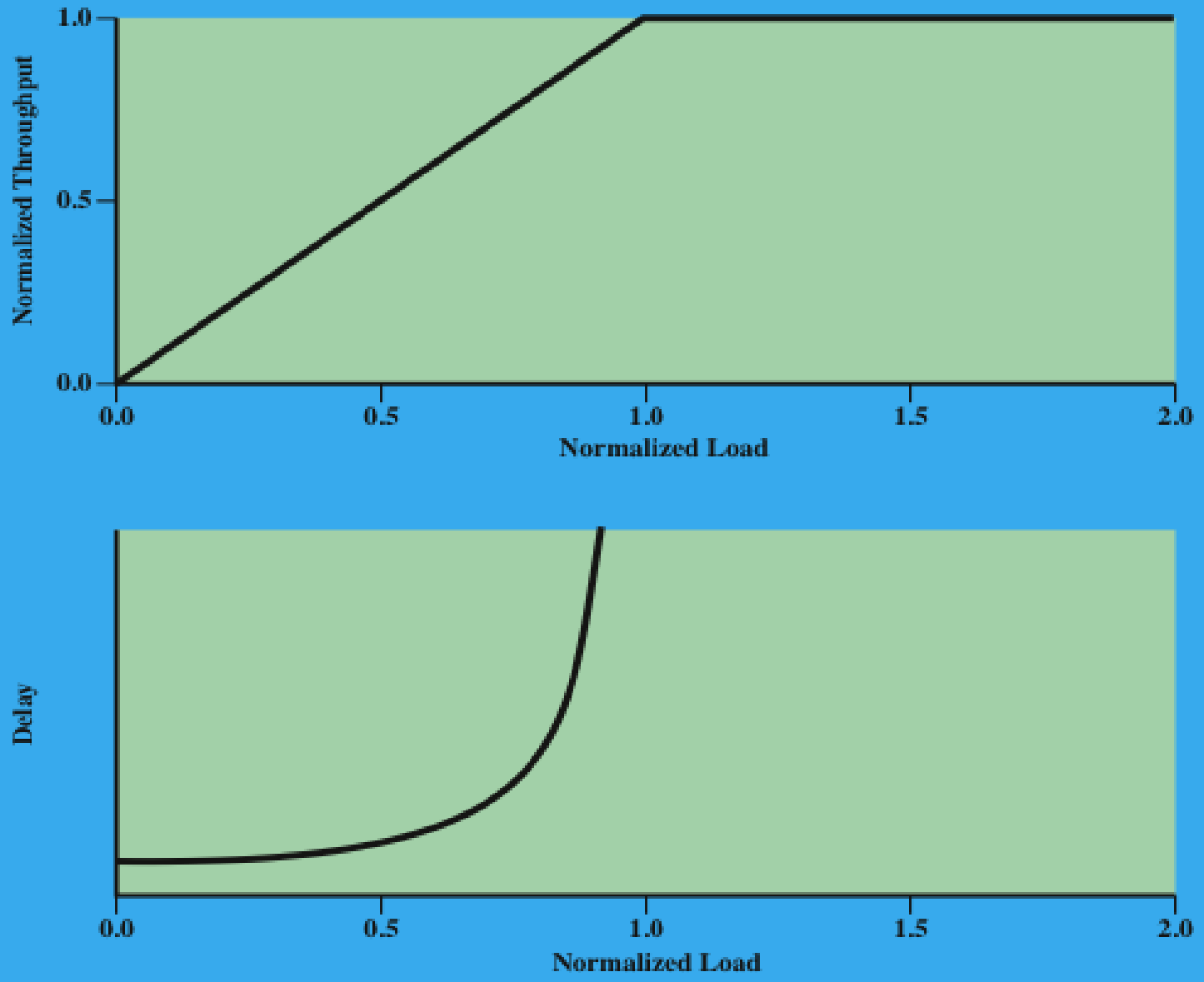


Figure 2.12 Ideal Network Utilization

Congestion Control

Consider what happens in a network with finite buffers if no congestion control or to restrain input from end systems.

- At light loads, throughput and hence network utilization increases as the offered load increases.

- As the load continues to increase, a point is reached beyond which the throughput increases at a rate slower than the rate at which offered load is increased.
  - network entry into a moderate congestion state.
  - The network continues to cope with the load, although with increased delays.
    - The load is not spread uniformly throughout the network. Therefore, while some nodes may experience moderate congestion, others experiencing severe congestion and may need to discard traffic.
    - In addition, to balance the load by routing packets through areas of lower congestion. For the routing function to work, an increased number of routing messages must be exchanged between nodes to alert congestion; this overhead reduces the capacity available for data packets.

- A point is reached beyond which throughput actually drops with increased offered load.
  - The reason for this is that the buffers at each node are of finite size.
  - When the buffers are full, the node must discard packets. Therefore, the sources must retransmit the discarded packets in addition to new packets.
    - More and more packets are retransmitted, the load on the system grows, and more buffers become saturated.
    - Even successfully delivered packets may be retransmitted because it takes too long, at a higher layer (for example, transport layer), and the sender assumes the packet did not get through and retransmits.
      - Under these circumstances, the effective capacity of the system declines to zero.

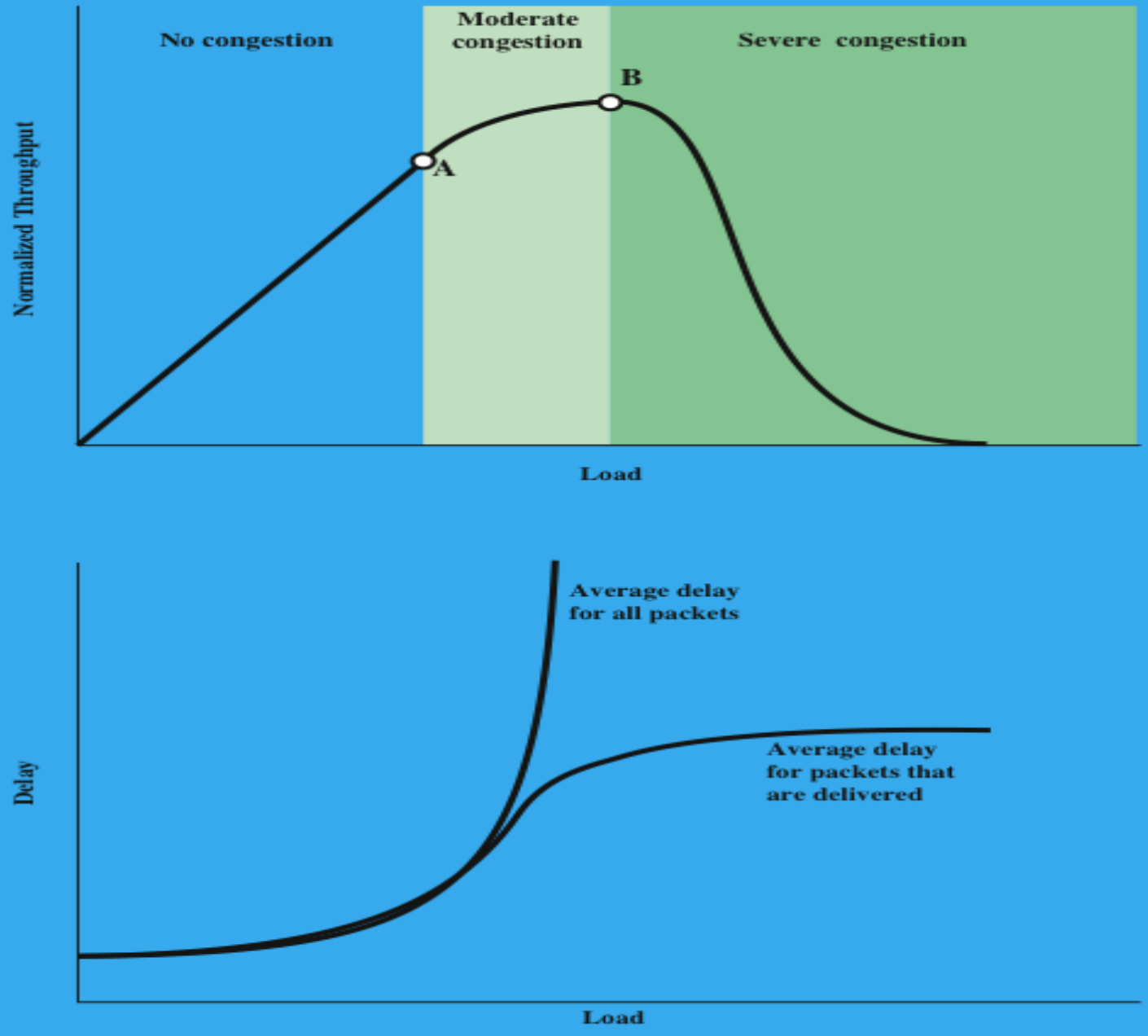


Figure 2.13 The Effects of Congestion

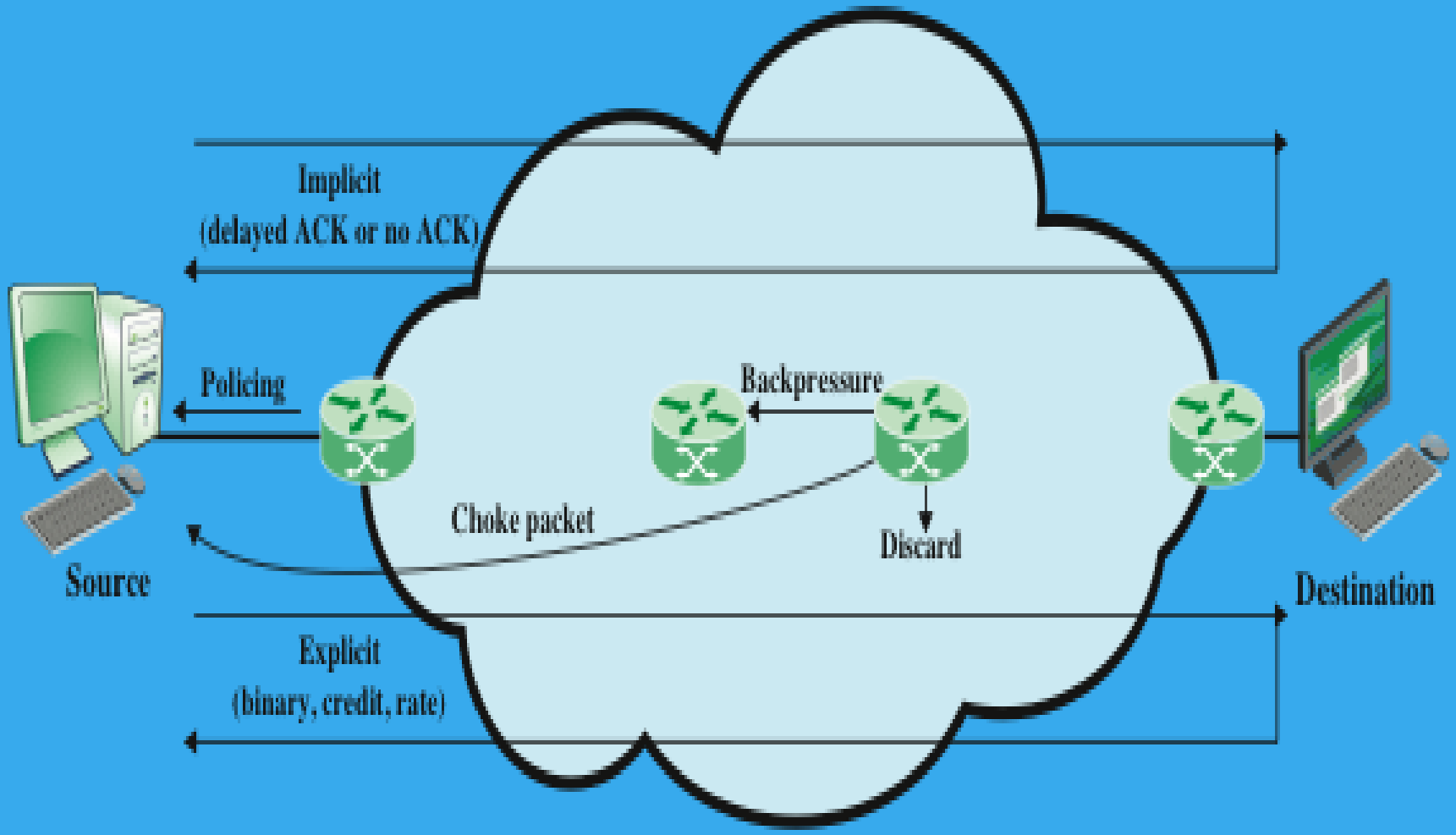


Figure: Mechanism of Congestion Control

Backpressure

- Can be exerted on the basis of links or logical connections
- Flow restriction propagates backward (against the flow of data traffic) to sources, which are restricted in the flow of new packets into the network
  - Backpressure for all traffic on a particular link is automatically invoked by the flow control mechanisms of data link layer protocols
  - Backpressure can also be selectively applied to logical connections so that the flow from one node to the next is only restricted or halted on some connections, generally the ones with the most traffic
  - Used in Frame Relay and Asynchronous Transfer Mode (ATM) networks
    - Use of these networks has declined considerably in favor of Ethernet carrier networks and IP-based Multiprotocol Label Switching (MPLS) networks

Choke Packet

- A control packet generated at a congested node and transmitted back to a source node to restrict traffic flow
- Either a router or a destination end system may send this message to a source end system
- On receipt of a choke packet the source host should cut back the rate at which it is sending traffic to the specified destination until it no longer receives choke packets

Choke Packet

- The choke packet can be used by a router or host that must discard IP datagrams because of a full buffer
  - The router will issue a choke packet for every packet that it discards
  - A system may anticipate congestion and issue choke packets when its buffers approach capacity
- Receipt of a choke packet does not imply delivery or nondelivery

Implicit Congestion Signaling

- When network congestion occurs, two things may happen:
  - The transmission delay for an individual packet from source to destination increases, so that it is noticeably longer than the fixed propagation delay
  - Packets are discarded
- If a source can detect increased delays and packet discards, it has implicit evidence of network congestion

Implicit Congestion Signaling

- Congestion control on the basis of implicit signaling
  - It is the responsibility of end systems
  - does not require action on the part of network nodes
- It is an effective congestion control technique in connectionless, or datagram, networks, such as IP-based internets

Explicit Congestion Signaling

- Explicit congestion signaling approaches can work in one of two directions:
  - Backward
    - Notifies the source that congestion avoidance procedures should be initiated where applicable for traffic in the opposite direction of the received notification
    - It indicates that the packets that the user transmits on this logical connection may encounter congested resources
    - Backward information is transmitted either by altering bits in a header of a data packet headed for the source or by transmitting separate control packets to the source

Forward

- Notifies the user that congestion avoidance procedures should be initiated where applicable for traffic in the same direction as the received notification
- It indicates that this packet, on this logical connection, has encountered congested resources
- In some schemes, when a forward signal is received by an end system, it echoes the signal back along the logical connection to the source;
- In other schemes, the end system is expected to exercise flow control upon the source end system at a higher layer

Categories of explicit congestion signaling approaches:

Binary

- A bit is set in a data packet as it is forwarded by the congested node; when a source receives a binary indication of congestion on a logical connection, it may reduce its traffic flow

Categories of explicit congestion signaling approaches:

Credit based

- These schemes are based on providing an explicit credit to a source over a logical connection; the credit indicates how many octets or how many packets the source may transmit
- Common for end-to-end flow control, in which a destination system uses credit to prevent the source from overflowing the destination buffers
- Defined in Frame Relay and ATM networks

Categories of explicit congestion signaling approaches:

Rate based

- These schemes are based on providing an explicit data rate limit to the source over a logical connection
- The source may transmit data at a rate up to the set limit
- To control congestion, any node along the path of the connection can reduce the data rate limit in a control message to the source

SDN and NFV

- Ever-increasing volume and variety of network traffic from sources as big data, cloud computing and mobile traffic
  - Difficult to meet QoS and QoE requirements
  - Networks need to be more adaptable and scalable
    - These two key technologies are being deployed largely by network service and application providers

Software-Defined Networking (SDN)

- Enhanced level of flexibility and customizability to meet the needs of newer networking and IT trends such as cloud, mobility, video etc.

- The two elements involved in forwarding packets through routers are
  - a control function:
    - decides the route the traffic takes and the relative priority of traffic.
  - a data function:
    - forwards data based on control-function policy.

Software-Defined Networking (SDN)

- Prior to SDN, these functions were performed at each network device (router, bridge, packet switch, and so on).
  - This approach is relatively inflexible
  - Requires all the network nodes to implement the same protocols.

Software-Defined Networking (SDN)

- A central controller performs all complex functionality, including routing, policy declaration, etc.
  - SDN control plane consists of one or more SDN controllers.
  - The controller defines the data flows that occur in the SDN data plane (The switches constitute the data plane).
  - Each flow is configured by the controller.
  - Controller computes a route for the flow to take, and adds an entry for that flow in each of the switches along the path.
    - Switches simply manage flow tables whose entries are populated by the controller.
    - Communication between the controller and the switches uses a standardized protocol.

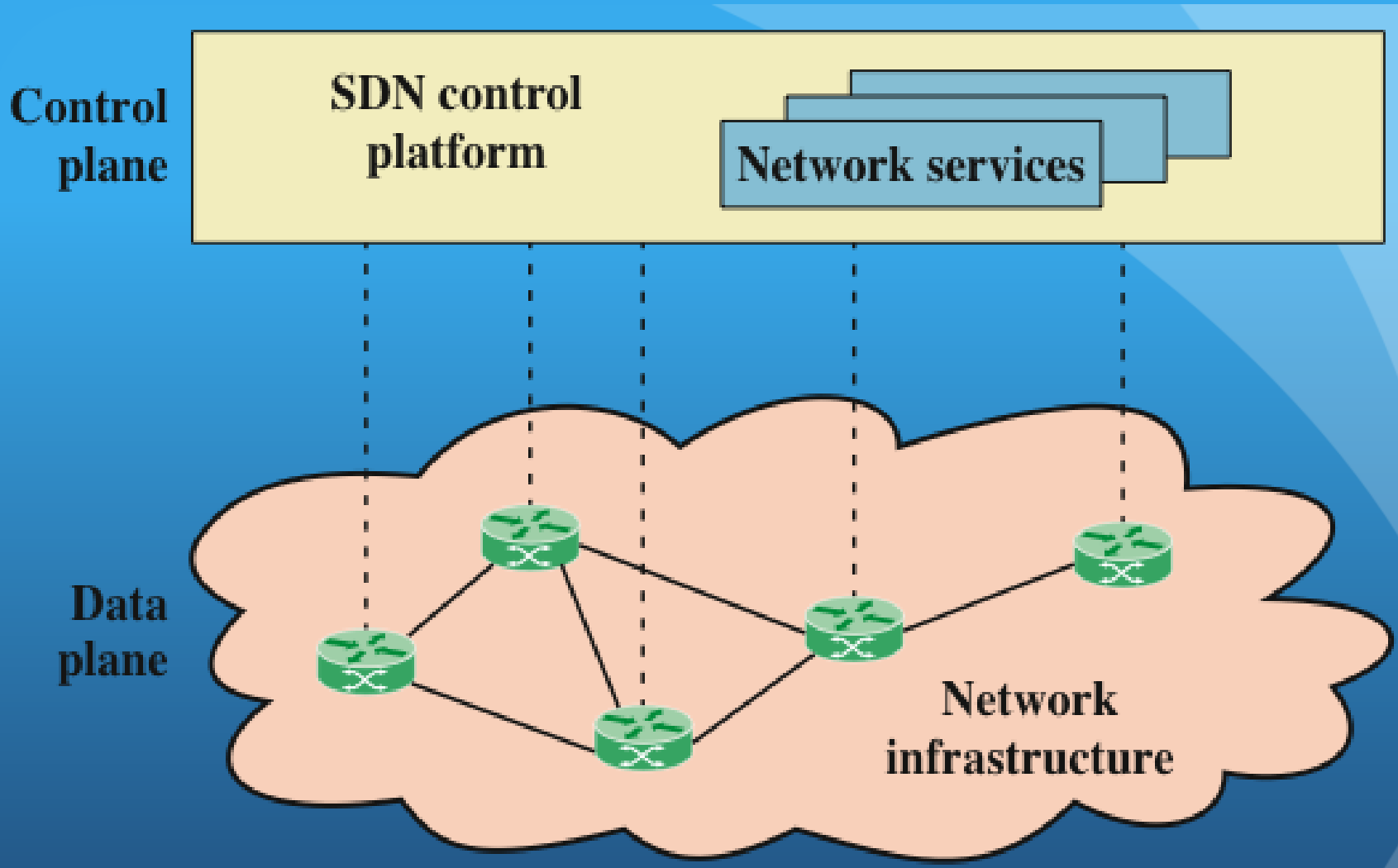


Figure 2.15 Software-Defined Networking

Key Drivers of SDN

- A driving factor for SDN is the increasingly widespread use of server virtualization
- Server virtualization masks server resources from server users
  - Partition a single machine into multiple, independent servers, conserving hardware resources
  - Possible to quickly migrate a server from one machine to another for load balancing or for dynamic switchover in the case of machine failure
  - It has become a central element in dealing with big data applications and in implementing cloud computing infrastructures

Key Drivers of SDN

- But virtualization creates problems with traditional network architectures.
  - One problem is configuring virtual LANs.
  - Network managers need to make sure the VLAN used by the virtual machine (VM) is assigned to the same switch port as the physical server running the VM.
  - But with the VM being movable, it is necessary to reconfigure the VLAN every time that a virtual server is moved.
    - This is difficult to do with conventional network switches, in which the control logic for each switch is collocated with the switching logic.

Key Drivers of SDN

- With server virtualization
  - A lot of traffic happens between virtual servers themselves (like syncing databases or handling security).
- Since these server-to-server connections can shift and grow over time
  - The network must be flexible enough to handle changing traffic patterns.

Key Drivers of SDN

- Employees now use smartphones, tablets, and laptops to access company resources, which can suddenly create heavy and unpredictable network traffic.
- Since devices move and connect from different places
  - network managers must quickly adjust resources, security, and quality of service.

Key Drivers of SDN

- Current networks can handle different QoS and security needs for traffic, but the setup takes a lot of time in big enterprises.
- Network managers must configure each vendor's devices separately and fine-tune settings for each session or application.
  - As a result, bringing up a new VM can require hours or even days of reconfiguration.

Network Functions Virtualization

- VM technology been used, until recently, for application-level server functions such as database servers, cloud servers, web servers, e-mail servers, and so on.
- This VM technology can equally be applied to network devices, such as routers, LAN switches, firewalls, and IDS/IPS servers.

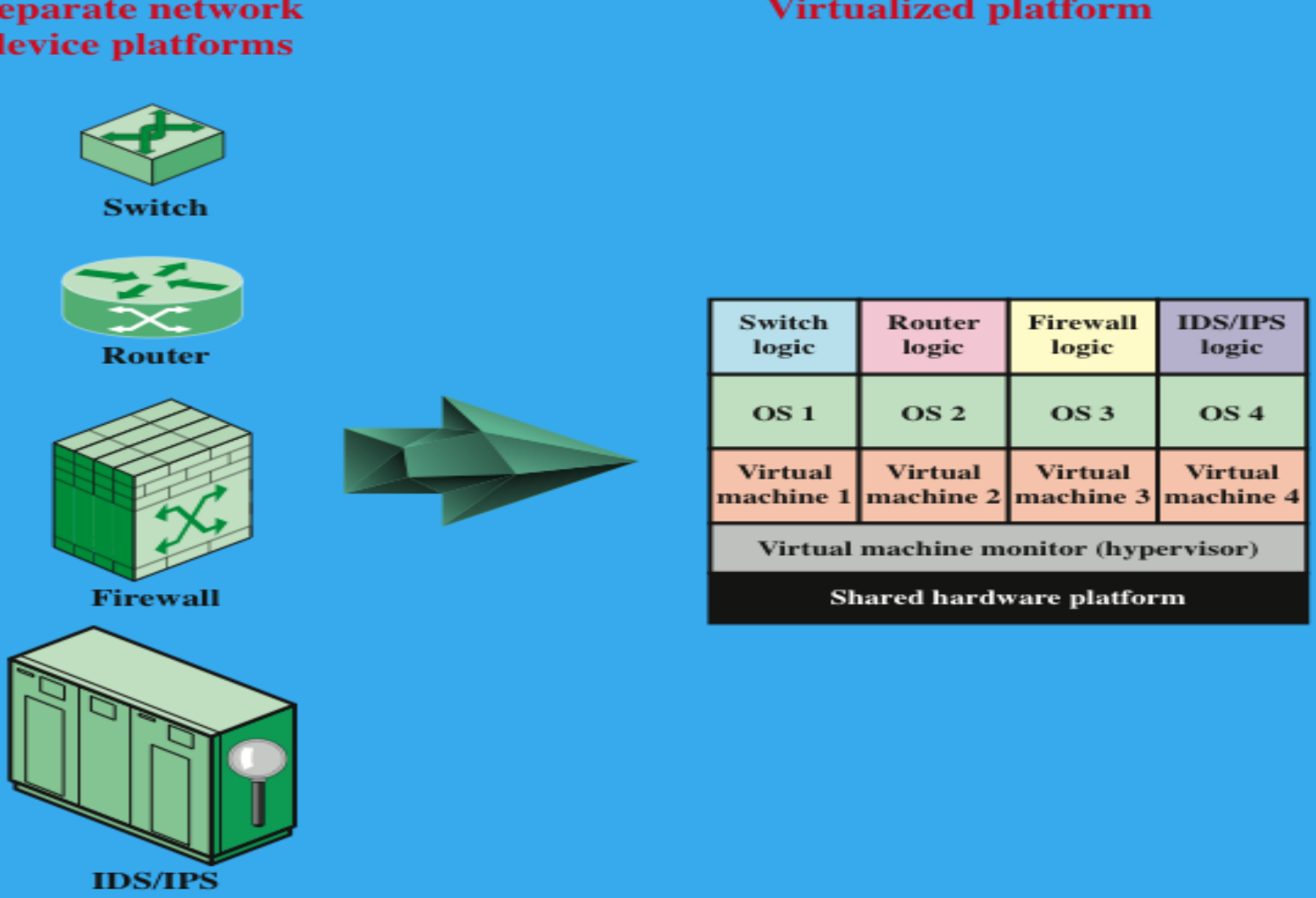


Figure 2.16 Network Functions Virtualization

Network Functions Virtualization (NFV)

- NFV decouples network functions from proprietary hardware platforms and implements these functions in software.
  - routing
  - firewalling
  - intrusion detection
  - Network Address Translation
- It utilizes standard virtualization technologies that run on high-performance hardware to virtualize network functions.
- It is applicable to any data plane processing or control plane function in both wired and wireless network infrastructures.

Network Functions Virtualization

- NFV has a number of features in common with SDN.
- They share the following objectives:
  - Move functionality to software
  - Use commodity hardware platforms instead of proprietary platforms
  - Use standardized or open application program interfaces (APIs)
  - Support more efficient evolution, deployment, and repositioning of network Functions

Network Functions Virtualization

- NFV and SDN are independent but complementary schemes.
- SDN decouples the data and control planes of network traffic control, making the control and routing of network traffic more flexible and efficient.
- NFV decouples network functions from specific hardware platforms via virtualization to make the provision of these functions more efficient and flexible.
- Virtualization can be applied to the data plane functions of the routers and other network functions, including SDN controller functions.
- So, either can be used alone, but the two can be combined to reap greater benefits.

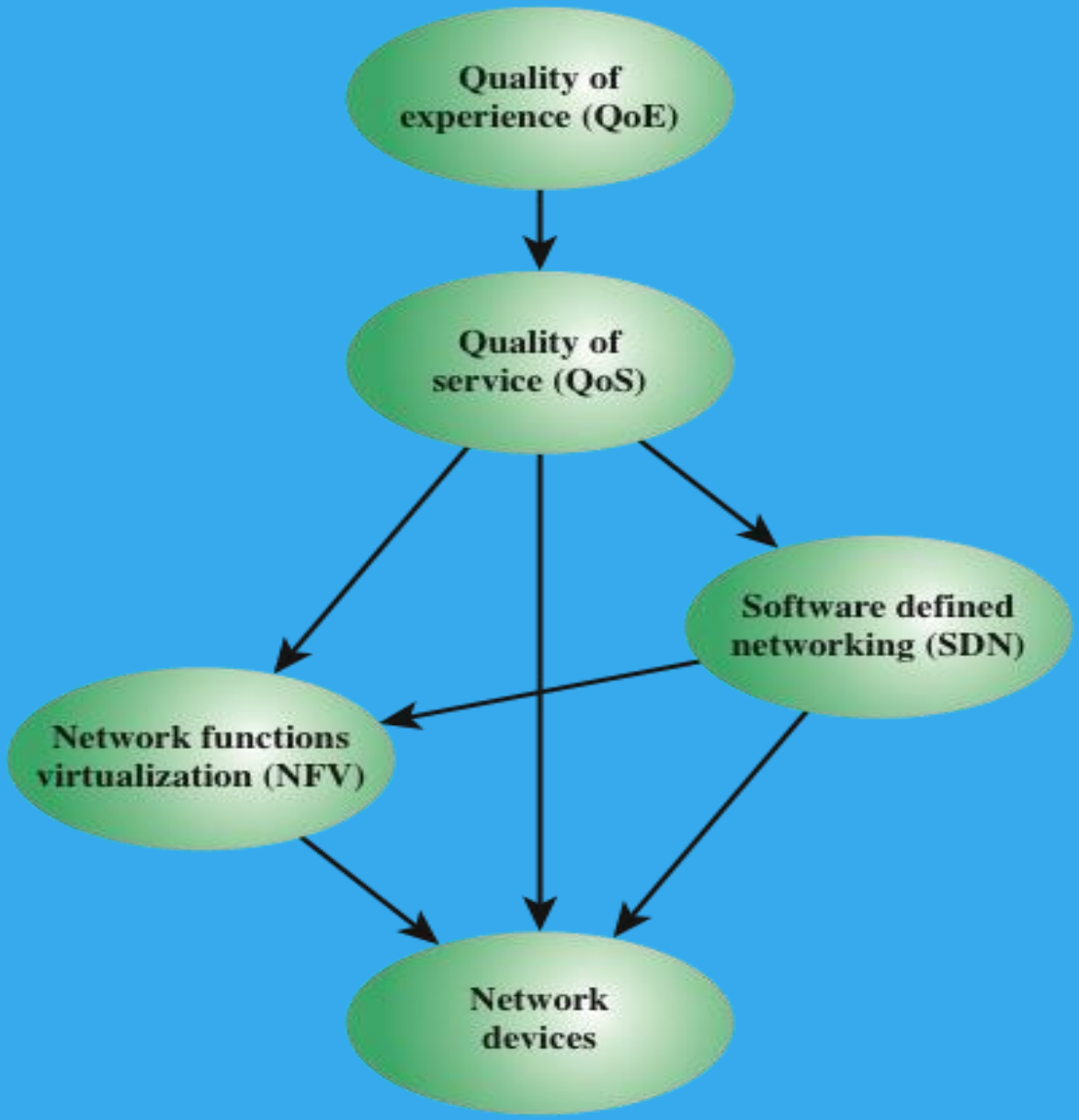


Figure 2.17 Modern Networking Schema

- Ultimately, the concern of a network service provider is about the set of network devices (such as routers) and the control and management of the functions they perform (such as packet forwarding).
- If NFV is used, network functions are implemented in software and executed on VMs.
- If instead the network functions are implemented on dedicated machines and SDN is used,
  - The control functions are implemented on central SDN controllers, which interact with the network devices.
- However, SDN and NFV are not mutually exclusive.

- If both SDN and NFV are implemented for a network, the following relationships hold:
  - Network data plane functionality is implemented on VMs.
  - The control plane functionality may be implemented on a dedicated SDN platform or on an SDN VM.
- In either case, the SDN controller interacts with the data plane functions running on VMs.
- QoS measures are commonly used to specify the service required by various network customers or users and to dictate the traffic management policies used on the network.

- The common case, until recently, is that QoS was implemented on network that used neither NFV nor SDN. In this case, routing and traffic control policies must be configured directly on network devices using a variety of automated and manual techniques.
- If NFV but not SDN is implemented, the QoS settings are communicated to the VMs. With SDN, regardless of whether NFV is used, it is the SDN controller that is responsible for enforcing QoS parameters for the various network users.
- If QoE considerations come into play, these are used to adjust QoS parameters tosatisfy the users' QoE requirements.

End of Chapter 2

Foundations of Modern Networking

SDN, NFV, QoE, IoT, and Cloud

By: William Stallings

Chapter 3

SDN: Background and Motivation

Evolving Network Requirements

- A number of Trends are driving network providers and users to reevaluate traditional approaches to network architecture.
- These trends are grouped under the categories
  - Demand
  - Supply
  - Traffic patterns

Evolving Network Requirements

- Demand is increasing
  - Cloud computing: A dramatic shift by enterprises to both public and private cloud services
  - Big data
    - The processing of huge data sets requires massive parallel processing on thousands of servers
      - Growing demand for network capacity
  - Mobile traffic
    - Employees are increasingly accessing enterprise network resources via personal devices
  - The Internet of Things (IoT): Generate modest traffic, sheer number of such devices results in a significant load on the enterprise network

Evolving Network Requirements

- Supply is increasing
  - Elements of modern networking technologies, WiFi and Ethernet, are well into the gigabits per second
  - 4G and 5G for mobile devices
  - The increase in the capacity of the network transmission technologies matches the performance of network devices, such as LAN switches, routers, firewalls, network monitoring systems etc.
  - Year by year, these devices have larger, faster memories, enabling greater buffer capacity and faster buffer access, as well as faster processor speeds.

Evolving Network Requirements

- Traffic patterns
  - As traffic patterns have changed and become more complex
    - Traditional enterprise network architectures are increasingly ill suited to the demand
    - Network architectures consists of Ethernet switches with routers connecting large Ethernets and connecting to the Internet
    - Well suited to client-server model
      - Traffic was mostly between one client and one server
      - Relatively predictable traffic volumes

- Traffic patterns (Cont…)
  - Reasons for more dynamic and complex traffic patterns within the enterprise data center, enterprise networks, and carrier networks
    - Client/server applications typically access multiple databases and servers
      - generating “horizontal” traffic between servers and “vertical” traffic between servers and clients.
    - Network convergence of voice, data, and video traffic creates unpredictable traffic patterns, often of large multimedia data transfers.
    - Unified communications (UC) strategies involve heavy use of applications that trigger access to multiple servers.

- Traffic patterns (Cont…)
  - Personal bring your own device (BYOD) policies, results in user access to corporate content and applications from any device anywhere any time.
    - an increasingly significant fraction of enterprise network traffic.
  - The widespread use of public clouds
    - Local traffic onto WANs for many enterprises, unpredicted load on enterprise routers
  - The now-common practice of application and database server virtualization has significantly increased the number of hosts requiring high-volume network access

Traditional Network Architectures are Inadequate

- As QoS and QoE requirements imposed on the network are expanded as a result of the variety of applications,
  - The traffic load must be handled in an increasingly sophisticated and agile fashion

Traditional Network Architectures are Inadequate

- The traditional internetworking approach is based on the TCP/IP protocol architecture;
  - Two-level end system addressing
    - Ethernet MAC addresses, a hardware based identifier
    - At the internetworking level, IP addresses
  - Routing based on destination
    - Datagram approach, successive packets find minimum delay path
    - Recently, satisfy QoS requirements, packets are often treated in terms of flows of packets.
  - Distributed, autonomous control approache developed
    - Predominantly static networks and end systems were fixed

Limitations

- The Open Networking Foundation (ONF) cites four general limitations of traditional network architectures:
  - Static, complex architecture
    - Traditional networks have become complex due to diverse requirements like QoS, fluctuating traffic, and security, leading to many independent protocols
    - Managing changes (e.g., adding or moving devices) requires manual reconfiguration across multiple devices—switches, routers, VLANs, and QoS settings
    - Adjusting QoS or policies for new traffic patterns is labor-intensive since admins must configure each vendor's equipment separately, often on a per-application or per-session basis.

Limitations

- Inconsistent policies
  - Applying a single security policy in a large network often requires reconfiguring thousands of devices.
  - When a new virtual machine starts, updating ACLs across the network can take hours or even days.
- Inability to scale
  - Network demand is growing fast, but adding capacity with multi-vendor equipment is difficult due to complex and static networks.
  - Enterprises often oversubscribe network links based on predicted traffic, but virtualization and diverse multimedia apps make traffic patterns unpredictable.

Limitations

- Vendor dependence
  - Enterprises need to launch new services quickly to meet changing demands.
  - Closed vendor equipment slows them down due to limited open interfaces.

The SDN Approach

- Requirements:

Adaptability	• Networks must adjust and respond dynamically, based on application needs, business policy, and network conditions
Automation	• Policy changes must be automatically propagated so that manual work and errors can be reduced
Maintainability	• Introduction of new features and capabilities must be seamless with minimal disruption of operations
Model management	• Network management let you change whole network at once, instead of configuring each device individually
Mobility	• Control functionality must accommodate mobility, including mobile user devices and virtual servers
Integrated security	• Network applications must integrate seamless security as a core service instead of as an add-on solution
On-demand scaling	• the ability to scale up or scale down the network and its services to support on-demand requests

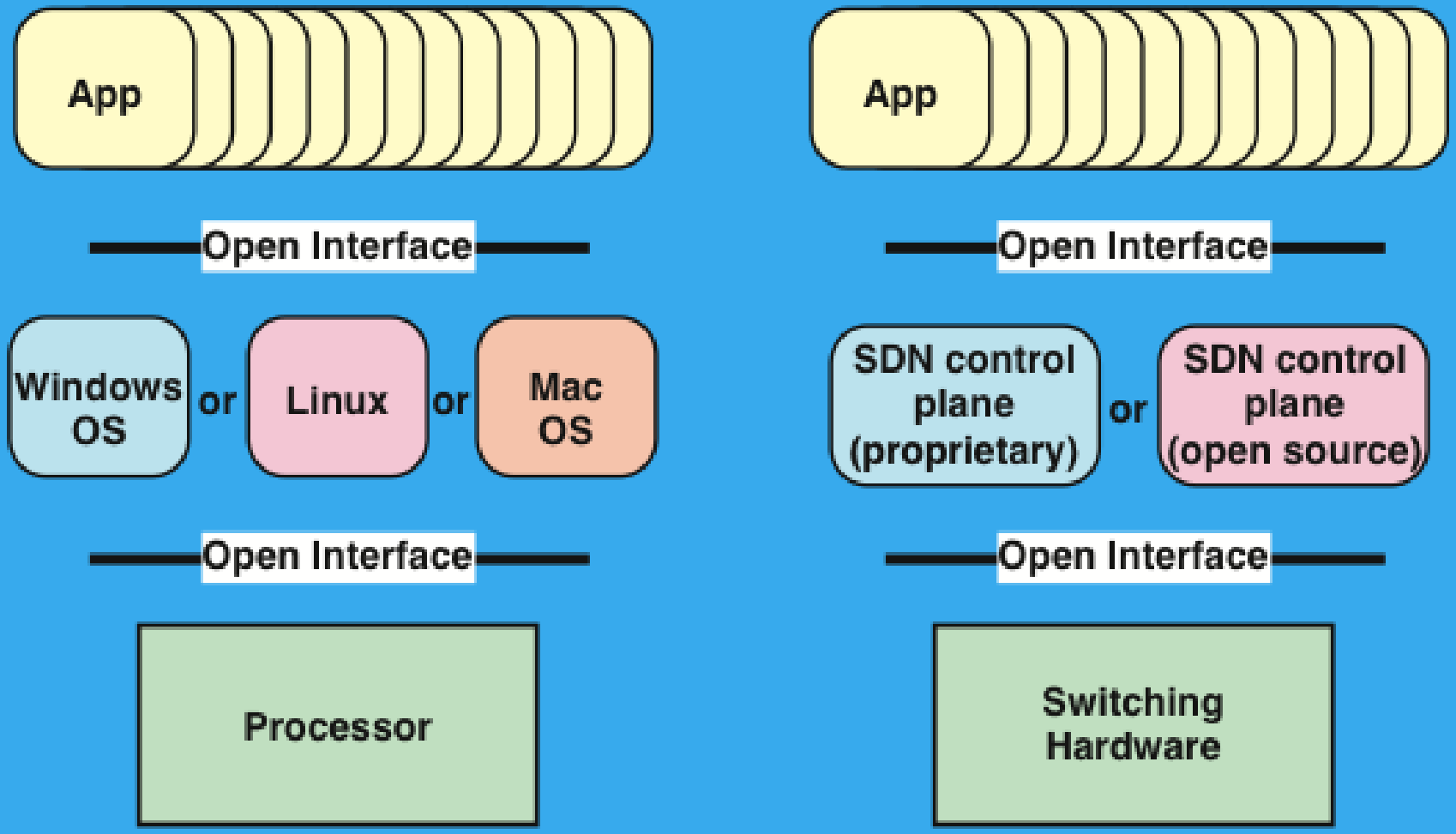


Figure 3.1 The Modern Approach to Computing and Networking

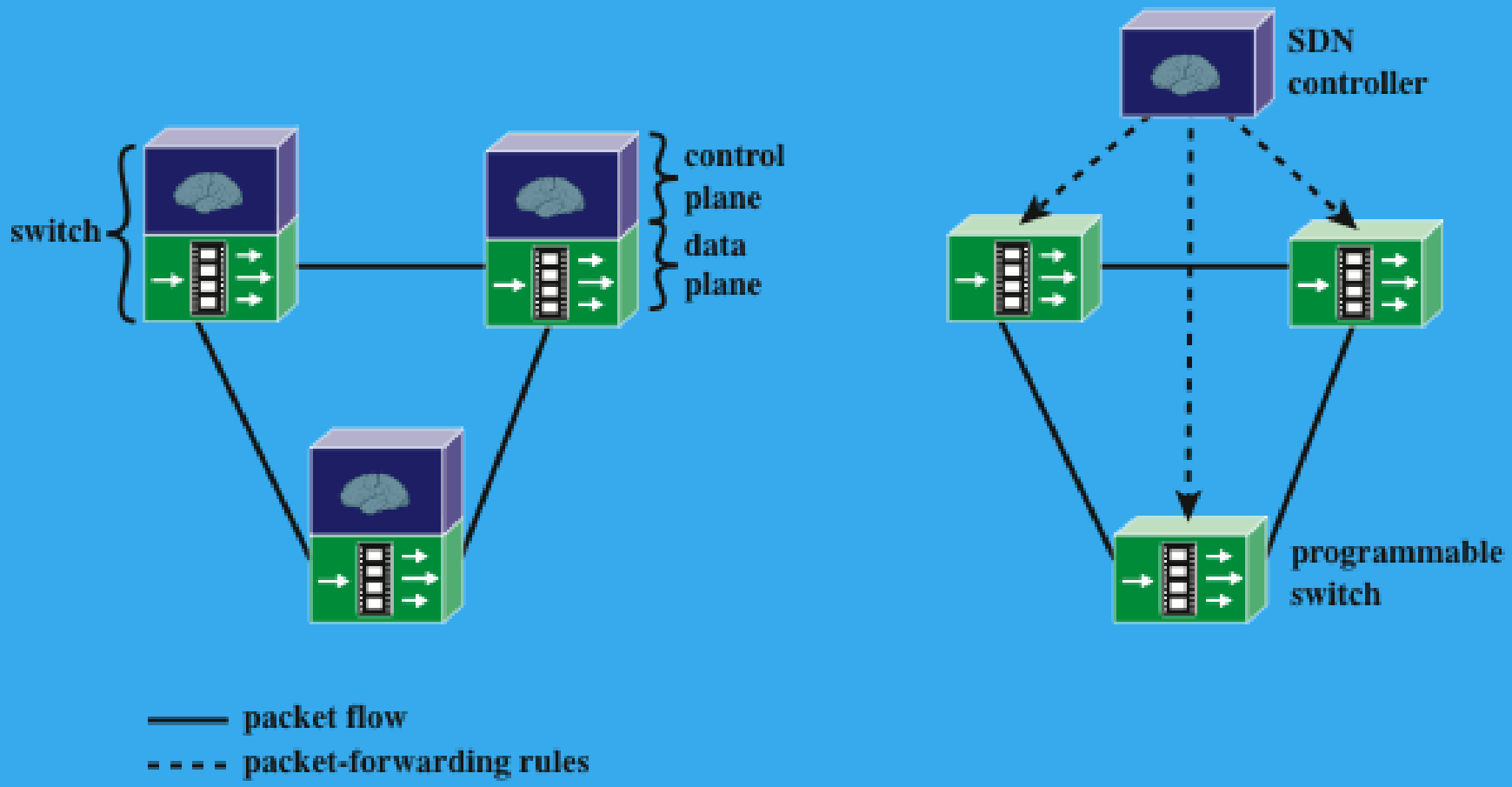


Figure 3.2 Control and Data Planes

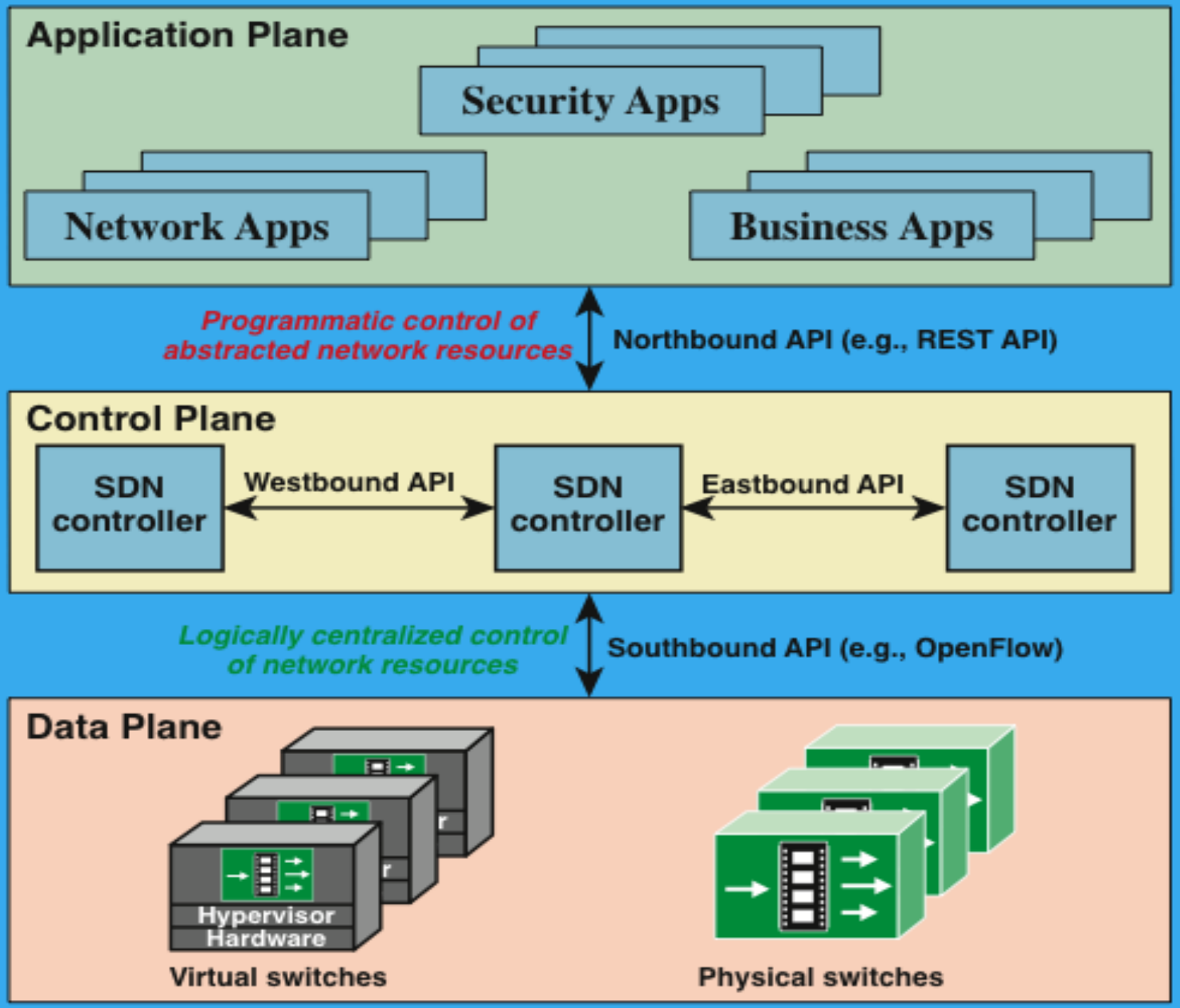


Figure 3.3 SDN Architecture

Characteristics of Software-Defined Networking

- The control plane is separated from the data plane; data plane devices become simple packet-forwarding devices
- The control plane is implemented in a centralized controller or set of coordinated centralized controllers
  - The SDN controller has a centralized view of the network or networks under its control
  - The controller is portable software that can run on commodity servers and is capable of programming the forwarding devices based on a centralized view of the network

Characteristics of Software-Defined Networking

- Open interfaces are defined between the devices in the control plane (controllers) and those in the data plane
- The network is programmable by applications running on top of the SDN controllers;
  - the SDN controllers present an abstract view of network resources to the applications

Organization	Mission	SDN-related effort
Open Networking Foundation (ONF)	An industry consortium dedicated to the promotion and adoption of SDN through open standards development.	OpenFlow
Internet Engineering Task Force (IETF)	The Internet's technical standards body. Produces RFCs and Internet Standards.	Interface to Routing Systems (I2RS) Service Function Chaining
European Telecommunications Standards Institute (ETSI)	An EU-sponsored standards organization that produces globally-applicable standards for information and communications technologies.	NFV Architecture
OpenDaylight	A collaborative project under the auspices of the Linux Foundation	OpenDaylight
International Telecommunication Union—Telecommunication Standardization Sector (ITU-T)	United Nations agency that produces Recommendations with a view to standardizing telecommunications on a worldwide basis.	SDN functional requirements and architecture
Internet Research Task Force (IRTF) Software Defined Networking Research Group (SDNRG)	Research group within IRTF. Produces SDN-related RFCs.	SDN architecture
Broadband Forum (BBF)	Industry consortium developing broadband packet networking specifications.	Requirements and framework for SDN in telecommunications broadband networks
Metro Ethernet Forum (MEF)	Industry consortium that promotes the use of Ethernet for metropolitan and wide area applications.	Defining APIs for service orchestration over SDN and NFV
IEEE 802	An IEEE committee responsible for developing standards for local area networks (LANs)	Standardize SDN capabilities on access networks.
Optical Internetworking Forum (OIF)	Industry consortium promoting development and deployment of interoperable networking solutions and services) for optical networking products	Requirements on Transport Networks in SDN Architectures

SDN and NFV Standards Activities

Organization	Mission	SDN-related effort
Open Data Center Alliance (ODCA)	Consortium of leading IT organizations developing interoperable solutions and services for cloud computing.	SDN Usage Model
Alliance for Telecommunications Industry Solutions (ATIS)	A standards organization that develops standards for the unified communications (UC) industry.	Operational Opportunities and Challenges of SDN/NFV Programmable Infrastructure
Open Platform for NFV (OPNFV)	An open source project focused on accelerating the evolution of NFV.	NFV intrastructure

SDN and NFV Standards Activities

Standards-Developing Organizations

- **Internet Society (ISOC)**
  - IETF and IRTF are the active groups
  - The coordinating committee for Internet design, engineering, and management
  - Areas covered include
    - The operation of the Internet itself
    - The standardization of protocols used by end systems on the Internet for interoperability
  - Various organizations under the ISOC
    - Responsible for the actual work of standards development and publication

Internet Engineering Task Force (IETF)

- Has working groups developing SDN-related specifications in the following areas:

Interface to routing systems (I2RS)

Develop capabilities to interact with routers and routing protocols to apply routing policies

Service function chaining

Develop an architecture and capabilities for controllers to direct subsets of traffic across the network in such a way that each virtual service platform sees only the traffic it must work with

Internet Research Task Force (IRTF)

- It has published Software-Defined Networking (SDN): Layers and Architecture Terminology (RFC 7426, 2015)
  - Request for Comments (RFC) provides discussion of the southbound API, describes APIs for I2RS
  - Sponsors SDN Research Group (SDNRG)
    - This group investigates SDN from various perspectives
    - Identifying the approaches that can be defined, deployed and used
    - Identifying future challenges

International Telecommunications Union–Telecommunication Standardization (ITU-T)

- A UN agency that issues standards, called recommendations, in the telecommunications area
  - their only published contribution to SDN is Recommendation Y.3300 (*Framework of Software-Defined Networking*, June 2014)
    - Document addresses definitions, objectives (programmability, flexibility, efficiency etc.), high-level capabilities (programmability, traffic control, multi-tenant etc.), requirements (scalability, reliability, open-interfaces etc.) and high-level architecture of SDN
- Has established a Joint Coordination Activity on Software-Defined Networking (JCA-SDN) and began work on developing SDN-related activities

International Telecommunications Union–Telecommunication Standardization (ITU-T)

- Four ITU-T study groups are involved in SDN-related activities:
  - SG 13 (Future networks, including cloud computing, mobile, and next-generation networks)
    - Group is studying SDN and virtualization aspects for next generation networks (NGNs)
  - SG 11 (Signaling requirements, protocols, and test specifications)
    - Studying the framework for SDN Signaling and how to apply SDN technologies for IPv6

International Telecommunications Union–Telecommunication Standardization (ITU-T)

- SG 15 (Transport, access, and home)
  - Looks at optical transport networks, access networks, and home networks
  - Transports aspect of SDN, aligned with ONF's SDN Architecture
- SG 16 (Multimedia)
  - Evaluating OpenFlow as a protocol to control multimedia packet flows
  - is studying virtual content delivery networks.

European Telecommunications Standards Institute (ETSI)

- Recognized by the European Union as a European Standards Organization
  - Not-for-profit SDO has member organizations worldwide and its standards have international impact
  - Has taken the lead role in defining standards for NFV
- ETSI's Network Functions Virtualization (NFV) Industry Specification Group (ISG) began work in January 2013 and produced a first set of specifications in January 2015

European Telecommunications Standards Institute (ETSI)

- The 11 specifications include
  - an NFV's architecture,
  - infrastructure,
  - service quality metrics,
  - management and orchestration,
  - resiliency requirements, and
  - security guidance

Industry Consortia

- Most important consortium involved in SDN standardization is the Open Networking Foundation (ONF)
  - ONF is an industry consortium dedicated to the promotion and adoption of SDN through open standards development
  - Most important contribution to date Is the OpenFlow protocol and API
    - OpenFlow protocol is the first standard interface specifically designed for SDN
    - Enables networks to evolve by giving logically centralized control software the power to modify the behavior of network devices through a well-defined "forwarding instruction set"

Industry Consortia

- Open Data Center Alliance (ODCA) is a consortium of leading global IT organizations dedicated to accelerating adoption of interoperable solutions and services for cloud computing
  - Defining requirements for SDN and NFV cloud deployment through the development of usage models for SDN and NFV

Industry Consortia

- Alliance for Telecommunications Industry Solution (ATIS) is a membership organization
  - provides the tools necessary for the industry to identify standards, guidelines, and operating procedures that make the interoperability of existing and emerging telecommunications products and services possible

OpenDaylight

- An open source software activity under the auspices of the Linux foundation
- Its member companies provide resources to develop an SDN controller for a wide range of applications
- Is more in the nature of an open development initiative than a consortium
- Also supports network programmability via southbound protocols, a bunch of programmable network services, a collection of northbound APIs, and a set of applications

Open Platform for NFV (OPNFV)

- An open source project dedicated to accelerating the adoption of standardized NFV elements
- Will establish a carrier-grade, integrated, open source reference platform that industry peers will build together to advance the evolution of NFV and to ensure consistency, performance, and interoperability among multiple open source components
- Will work with upstream projects to coordinate continuous integration and testing while filling development gaps

OpenStack

- Is an open source software project that aims to produce an open source cloud operating system
- Provides multitenant Infrastructure as a Service (IaaS) and aims to meet the needs of public and private clouds regardless of size, by being simple to implement and massively scalable
- Neutron is dedicated for networking, it provides network as a service
- SDN technology is expected to contribute to its networking part, and to make the cloud operating system more efficient, flexible, and reliable

End of Chapter 3